

RESEARCH

Open Access



Stress and light spectral quality influence the transcriptome of a tomato crop on the International Space Station

Anirudha R. Dixit¹, Christina L. M. Khodadad², Jenny M. Schuster³, LaShelle E. Spencer², Mary E. Hummerick², Robert C. Morrow⁴, Cary A. Mitchell⁵, Raymond M. Wheeler⁶ and Gioia D. Massa^{6*}

Abstract

Background Spaceflight introduces unique environmental stresses that challenge plant growth and development, necessitating an understanding of molecular responses to enable sustainable agriculture for long-duration missions. This study investigated the transcriptomic responses of tomato (*Solanum lycopersicum* cv. 'Red Robin') grown under red- and blue-rich light spectra aboard the International Space Station (ISS) and in ground controls at Kennedy Space Center. Using RNA sequencing, we analyzed leaf and adventitious root tissues to uncover gene expression changes driven by spaceflight conditions and light quality.

Results Our results revealed significant transcriptomic alterations influenced by both spaceflight and lighting spectra. Differential gene expression analysis identified numerous up- and downregulated genes in flight samples compared to ground controls, with adventitious roots displaying pronounced transcriptional changes. Genes associated with stress responses, hormonal signaling, and metabolic pathways were prominently upregulated in spaceflight-grown tissues. Gene ontology and KEGG pathway enrichment analyses highlighted critical roles for oxidative stress response, secondary metabolite biosynthesis, and hormonal regulation. Additionally, red-rich lighting appeared to stabilize gene expression patterns, while blue-rich lighting induced greater variability across conditions. Notably, spaceflight exhibited extensive adventitious roots formation, with gene clusters enriched in antioxidant metabolism, cell wall remodeling, and stress adaptation processes. Leaf tissues demonstrated distinct transcriptional signatures under flight and ground conditions, with blue-rich lighting enhancing stress-responsive pathways and red-rich lighting favoring metabolic stability.

Conclusions This study provides novel insights into the molecular mechanisms underlying plant adaptation to spaceflight and variable lighting conditions, emphasizing the importance of tailored environmental control for optimizing crop production in space. These findings could have implications for developing resilient plant varieties suitable for extreme environments, both in extraterrestrial settings and on Earth.

Keywords Adventitious roots, Differentially expressed genes, Gene ontology, Light, Spaceflight, Space Crop Production, Transcriptome, Veggie

*Correspondence:
Gioia D. Massa
gioia.massa@nasa.gov

Full list of author information is available at the end of the article



© NASA Laboratory Support Services and Operations (LASSO) II Contract, Astrion, Sierra Space and Jenny M. Schuster, Cary A. Mitchell 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Background

Plants undergo complex adaptive responses to environmental stresses, both biotic (e.g., pathogens) and abiotic (e.g., light spectral quality, water availability), which lead to transcriptomic changes that reveal how plants adjust gene expression under stress [1–3]. Spaceflight, a relatively new frontier for agricultural research, introduces unique environmental stresses, such as altered gravitational forces, microgravity-induced fluid dynamics, and supra-optimal CO₂ levels, that complicate plant responses in ways not observed on Earth [4–7]. Understanding how these conditions impact plant growth and development is essential for enabling sustainable food production for long-duration space missions, where fresh produce could enhance astronauts' nutrition, psychological well-being, and overall mission sustainability [8, 9].

While past studies have investigated leafy greens like mizuna and lettuce to assess plant responses to spaceflight, they have largely focused either on vegetative growth or plant associated microbial communities as part of food safety analyses [10–12]. This leaves a gap in understanding how fruiting crops complete their full developmental cycle under spaceflight conditions. Fruiting crops such as tomato (*Solanum lycopersicum* cv. 'Red Robin') remain underexplored but are of particular interest due to their nutritional value and psychological benefits for crewed missions. Unlike leafy greens, tomatoes require successful completion of flowering, pollination, and fruit development stages that may be especially sensitive to spaceflight-induced stresses. One of the more intriguing developmental observations of tomatoes grown on the International Space Station (ISS) was extensive adventitious root formation. On Earth, adventitious roots emerge from stems under hypoxic conditions resulting from flooding, and reduced nutrient availability [13–15]. Adventitious roots typically form in response to hormone-signaling pathways involving auxin, ethylene, and cytokinin [10, 16]. The mechanisms driving tomato adventitious root formation under spaceflight conditions remain unclear. Observations of extensive adventitious root growth by ISS-grown tomatoes contrast sharply with much lower adventitious root growth by ground controls, indicating that unique spaceflight-induced stressors, potentially due to challenges in root zone caused by water distribution driven by root-zone oxygenation, may be at play [11].

Light spectra are fundamental to plant development, affecting photosynthesis, growth rates, and nutritional content [17, 18]. In Earth-based studies, combinations of red and blue light are particularly effective in promoting growth, while broad-band spectra, such as red-blue-green or red-blue-white, further enhance yield and elemental content in various crops, including tomatoes [19]. The ISS "Veggie" Vegetable Production Units utilize

LED arrays with red, blue, and green lights, enabling controlled experiments to examine influences of different light spectra on plant physiology during spaceflight [20]. Previous work has shown that lighting impacts the nutritional quality of crops on Earth and in space, making it a critical factor in cultivating viable crops for space missions [21–24]. Tomatoes show different growth responses and quality under red-rich or blue-rich lighting, providing a deeper understanding of how to design light recipes for optimal space crop production [25–27].

There is existing evidence that spaceflight can modify how plants respond to red and blue light, relative to their behavior on Earth. On Earth, red and blue light elicit distinct developmental and stress-regulatory pathways, but in microgravity, these light responses may be disrupted or amplified due to altered photoreceptor localization, impaired auxin transport, and disturbed cytoskeletal alignment [28, 29]. In microgravity experiments using the European Modular Cultivation System (EMCS) onboard the ISS, blue-light-driven phototropic curvature in both *Arabidopsis* hypocotyls and roots was significantly reduced compared to 1 g controls, indicating that phototropin-mediated growth responses are gravity-dependent. Additionally, red-light mediated phototropism, regulated by phytochromes, was also diminished under fractional gravity and microgravity, further supporting the notion that both phototropin and phytochrome pathways are impacted by reduced gravity. These findings suggest that the same spectral inputs (red or blue light) may produce distinct transcriptional outcomes in space due to altered integration with hormonal and spatial sensing pathways.

To our knowledge, this is one of the first studies to explore how tissue-specific gene expression in a fruiting crop responds to distinct light spectra under true microgravity conditions. While the primary goal of this study was to investigate how red- and blue-enriched light affect tomato gene expression in microgravity compared to Earth, the unique environment aboard the ISS introduced additional biological insights. Specifically, in the Veggie growth chamber, microgravity and reduced convection presented challenges for water distribution, leading to moisture accumulation near the shoot–root interface. These conditions, along with microgravity stress, could have promoted adventitious root formation. Although we were unable to quantify the extent of root formation across treatments, we sampled these tissues to better understand how spaceflight-associated stressors interact with light-responsive gene networks. This inclusion provides a more comprehensive view of tissue-specific responses to light and environmental stress in space-grown tomato plants.

This study aimed to investigate the transcriptomic profiles of tomato plants subjected to spaceflight and

red- or blue-rich lighting conditions. We sampled leaves and adventitious roots from spaceflight-grown tomato plants under each light regime to elucidate the molecular pathways activated in response to spaceflight and modified lighting spectra compared to ground controls. Specifically, we sought to answer the following questions: (1) How does spaceflight alter gene expression in tomato leaves and adventitious roots? (2) Do red- and blue-rich lighting regimes modulate transcriptional responses differently under spaceflight versus ground conditions? (3) What stress-related molecular pathways are associated with the extensive adventitious root formation observed in space-grown plants? By addressing these questions, this study provides insight into plant adaptive responses under spaceflight and modified lighting spectra, offering implications for optimizing environmental conditions in bioregenerative life support systems.

Beyond space-based agriculture, findings from this research may improve terrestrial agricultural practices, particularly in challenging environments characterized by flooding, low oxygen levels, or nutrient-deficient conditions. This knowledge could contribute to the development of resilient crop varieties capable of surviving in challenging environments on Earth, such as those impacted by climate change and resource scarcity [12, 30, 31]. Consequently, this research contributes to our broader understanding of plant responses to extreme conditions, which has direct applications both in space and on Earth.

Methods

Plant growth conditions and spaceflight experiment setup
Tomato seeds (*Solanum lycopersicum* cv. ‘Red Robin’) were purchased from Totally Tomatoes (Randolph, WI, United States). Seeds were surface sanitized using a chlorine bleach-HCl method described by Massa et al., [32], except that hydrochloric acid volume was increased from 0.5 mL to 0.75 mL in 30 mL of household bleach (5.25% sodium hypochlorite). Briefly, seeds were exposed to chlorine gas for 1 h and then allowed to off-gas overnight

in a laminar flow hood. Seeds were tested for surface sanitization by placing up to 10 seeds on both tryptic soy agar (TSA) for detection of bacteria, and on inhibitory mold agar (IMA) for fungal and yeast detection. Plates were then allowed to incubate for 48–72 h. Acceptance criteria for seed surface sanitization and germination were 90% [32].

Veggie rooting packets (called “pillows”) were assembled as described by Massa et al. [32] using Profile Porous Ceramics (PPC) as a substrate. Substrate was mixed with controlled-release polymer-coated fertilizer in the following proportions per unit porous ceramic: Florikan Nutri-cote® 14–4–14 (T100, 4 g/L; T180, 6 g/L) with Florikan CRF (controlled-release fertilizers) 0–0–19 + 9% Magnesium (1 g/L; Florikan E.S.A., Sarasota, FL, United States) and Maxi Cal (1 g/L; CaCO₃; Kelly’s Green Team, Kirksville, MO, United States). ‘Red Robin’ seeds (three per pillow) were secured to wicks in each pillow with gum guar and sealed inside gas-impermeable bags for transfer to space. Plant pillows for this experiment, designated as VEG-05, arrived at the ISS aboard SpaceX CRS-26 on November 26, 2022. Before initiation of the VEG-05 experiment, light mapping was completed for each Veggie Production Unit using a LI-COR light meter (Lincoln, NE, United States) to measure photosynthetically active radiation (PAR measured as $\mu\text{mol m}^{-2} \text{s}^{-1}$) in the Veggie unit when lights were both on and off. The red, blue, and green light intensity at 10 cm distance from the light banks and an additional five positions across each Veggie unit were measured. The Veggie LED system includes red LEDs with a peak wavelength of 630 ± 10 nm, blue LEDs peaking at 455 ± 10 nm, and green LEDs peaking at 530 ± 10 nm, as described by Massa et al., [33]. The lighting set points to achieve the same PAR with each red-rich or blue-rich recipe were determined from these measurements (Table 1). Six (6) pillows were placed in Veggie unit 1 under a red-rich light regime while 6 pillows were placed in a second Veggie unit under a blue-rich light regime, with a 16/8 day/night photoperiod. Seed germination was initiated with water addition in the Vegetable

Table 1 Lighting setpoints and estimated PAR levels ($\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) for VEG-05 flight and ground conditions

Red-Rich					Blue-Rich			
Flight			Ground		Flight		Ground	
Locker	S/N001		S/N003		S/N010		S/N006	
Color	Setting	PAR*	Setting	PAR*	Setting	PAR*	Setting	PAR*
Red	260.00	228.13	220.00	225.11	150.00	125.43	120.00	126.85
Blue	40.00	29.09	30.00	27.93	150.00	125.36	130.00	127.84
Green	on (30)	22.50	on (30)	25.06	on (30)	26.58	on (30)	26.26
	Sum	279.72	Sum	278.09	Sum	277.37	Sum	280.96

*PAR is estimated average per Veggie

Production System on December 14, 2022. After germination, plants were grown for 100 days. Fruit were harvested at days 83, 90, and 100 as part of the growth experiment, but all RNA-seq tissue samples (leaves and stems with adventitious roots) were collected exclusively at the final harvest on day 100.

Veggie is open to the ISS environment, pulling air in from the space station Columbus module. The Veggie interior environment is then changed with heat from the lights and humidity from the plants. Inside the Veggie unit, environmental parameters (relative humidity and temperature) were recorded using a HOBO® data logger (HOBO Data Loggers, Bourne, MA, United States). ISS ambient conditions including temperature, relative humidity, and CO₂ levels were monitored by the ISS environmental sensors within the Columbus module. These environmental parameter measurements were used to set the target conditions (setpoints) for the International Space Station Environmental Simulator (ISSES) chambers on Earth at KSC. The actual ISSES chamber data represents the environmental values recorded during operation as the chambers worked to mimic the ISS setpoints. The identical Min–Max ranges observed for CO₂

and other environmental parameters from the ISS across both red-rich and blue-rich light conditions in flight and ground settings reflect the recorded ranges of the monitoring system, not sensor malfunction. All sensors were functional throughout the experiment (Table 2), however humidity sensors on HOBO® data loggers did not accurately represent air humidity as these sensors were at the level of plant pillows and became occluded with excess water under high moisture situations. Although the HOBO® data loggers recorded extremely high humidity values, these measurements were not considered reliable due to sensor saturation caused by the presence of free water on the surface. After the final harvest tissues were frozen at –80 °C in the Minus Eighty-Degree Laboratory Freezer for ISS (MELFI) and returned to Earth on April 15, 2023, aboard SpaceX CRS-27 and in June 2023 aboard SpaceX-CRS 28. Samples were maintained at –80 °C until further processing at Kennedy Space Center (KSC), FL. Ground controls were grown on Earth at KSC using the same fertilizer composition and blends of PPC in chambers mimicking environmental growth conditions to ISS. The final harvest of tomato plants on ISS was completed between 8:00 AM–12:00 PM (6:00–8:30 PM KSC time).

Table 2 Mean ($\pm$ SD) and range of environmental conditions (Min–Max) under red-rich and blue-rich light treatments for both flight and ground controls, as recorded by HOBO® data loggers (temperature, RH) and ISS environmental monitors (temperature, RH and CO₂). ISS ambient environmental parameters were used as setpoints for the ISSES ground-control chambers to mimic ISS conditions (excluding microgravity). HOBO® data loggers recorded temperature and relative humidity inside the Veggie units and reflect microenvironmental variations across lighting treatments and between flight and ground settings, however humidity data were not reliable due to saturation

	Data loggers/sensors	Red-rich		Blue-rich	
		Flight	Ground	Flight	Ground
		Mean $\pm$ SD (Min-Max)	Mean $\pm$ SD (Min-Max)	Mean $\pm$ SD (Min-Max)	Mean $\pm$ SD (Min-Max)
Temp (°C)	ISS Monitor	23.0 $\pm$ 0.4 (18.0–27.5)	22.9 $\pm$ 0.4 (16.3–25.7)	23.0 $\pm$ 0.4 (18.0–27.5)	22.9 $\pm$ 0.4 (16.3–25.7)
	HOBO®	23.8 $\pm$ 1.3 (20.1–26.3)	21.4 $\pm$ 0.9 (18.1–25.0)	23.5 $\pm$ 1.5 (19.2–28.1)	22.4 $\pm$ 1.2 (18.2–26.3)
RH (%)	ISS Monitor	37.7 $\pm$ 2.1 (31.0–58.0)	37.9 $\pm$ 2.3 (32.0–59.0)	37.7 $\pm$ 2.1 (31.0–58.0)	37.9 $\pm$ 2.3 (32.0–59.0)
	HOBO®	97.8 $\pm$ 7.1 (43.0–100.0)	82.4 $\pm$ 8.9 (37.4–98.1)	88.8 $\pm$ 6.9 (42.4–98.9)	93.9 $\pm$ 11.6 (37.4–100.0)
CO ₂ (ppm)	ISS Monitor	1946.6 $\pm$ 10.1 (1922.0–1965.0)	2010.6 $\pm$ 97.6 (919.0–4282.0)	1946.6 $\pm$ 10.1 (1922.0–1965.0)	2010.6 $\pm$ 97.6 (919.0–4282.0)

The plants were removed from Veggie and harvested on a tabletop under the ISS-cabin lights. Harvesting of the ground-control experiment at KSC was done under similar conditions.

RNA extractions and transcriptome sequencing

Plant tissue samples remained frozen at -80°C until further processing at KSC, FL. During flight, the tomato plants on the ISS developed an extensive adventitious root system (Fig. 1 A & B). However, the ground controls showed fairly typical tomato adventitious root growth (Fig. 1 C & D). These above-ground root systems were sampled from each plant, stored at -80°C , and returned to Earth for analysis. For flight-grown plants, total RNA was extracted from 5 biological replicates each of leaf and adventitious root tissues under red-rich light, and 4 biological replicates each under blue-rich light. For ground controls, 6 biological replicates were obtained for leaf and 3 biological replicates for adventitious root tissues grown under red-rich and blue-rich light, respectively. Leaf and adventitious root samples were ground to a fine powder in liquid nitrogen using mortar and pestle and stored at -80°C until further processing. Total RNA was isolated using the ZymoBIOMICS™ RNA Miniprep kit (ZYMO Research, Irvine, CA, United States) followed

by DNase-I treatment (NEB, MA, USA) and cleanup using RNeasy MinElute Cleanup Kit (QIAGEN, MD, USA). RNA was quantified using the Qubit™ RNA High Sensitivity (HS) assay kit (Invitrogen™, Grand Island, NY, United States), and quality was determined using a 2100 Bioanalyzer instrument (Agilent, Santa Clara, CA, United States). Sequencing libraries were prepared using Illumina's TruSeq Stranded mRNA library preparation kit along with Illumina TruSeq UD Indexes V2 dual indices following manufacturer's protocol without deviation. Concentration and quality of the pooled library for RNA sequencing was verified with Qubit™ RNA High Sensitivity (HS) assay kit and 2100 Bioanalyzer instrument, respectively. The mRNA library was sequenced using paired-end (2×150 bp) reads on the NovaSeq X Plus platform, generating approximately 2 billion reads at Admera Health, LLC (South Plainfield, NJ, United States).

Data processing and downstream analyses

RNA sequencing data were processed as described by Overbey et al., [34]. Briefly, raw sequencing reads were quality-trimmed using TrimGalore (ver. 0.6.10) with the following options: `-paired`, `-quality 30`, `-stranded_illumina`, and `-cores 4`, with all other parameters set to default. A Phred quality score cutoff of 30 was applied to

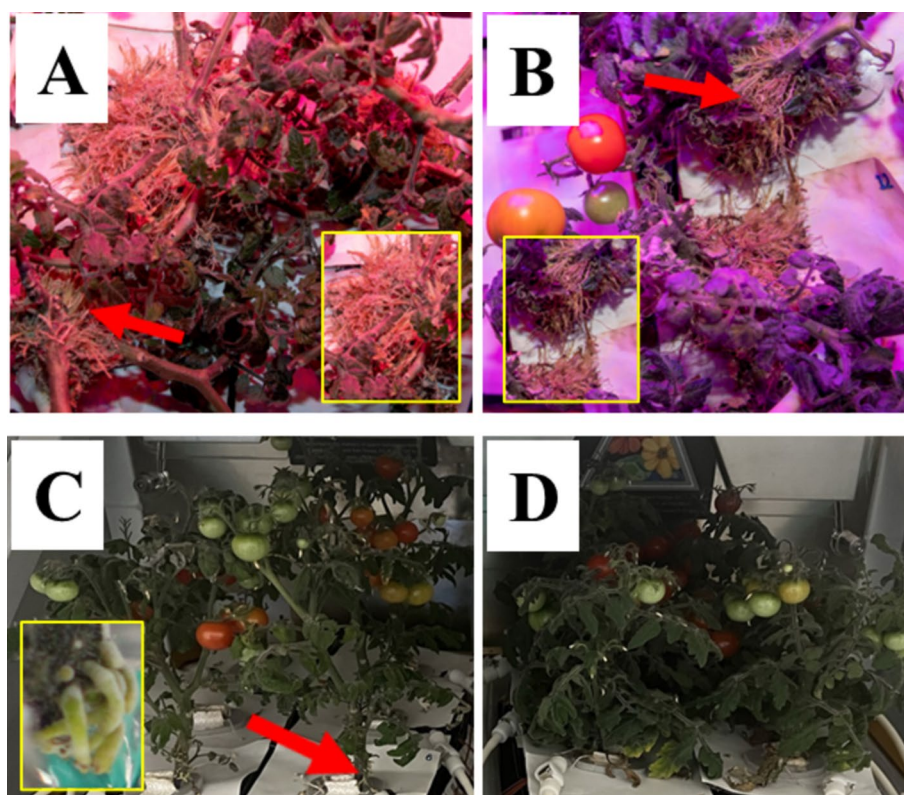


Fig. 1 Tomato plants grown on the International Space Station under red-rich light (A) and blue-rich light (B) capturing the extensive adventitious roots. The adventitious roots of ground controls plants grown under similar conditions at Kennedy Space Center under red-rich light (C) and blue-rich light (D) were not as evident

trim low-quality bases from the ends of reads. The default minimum read length threshold of 20 bp was used and reads shorter than this were discarded. For paired-end data, only read pairs in which both mates passed the quality and length filters were retained; if one read in a pair failed, the entire pair was discarded, as per TrimGalore's default behavior [35]. Using the STAR aligner [36] trimmed reads were aligned to the *Solanum lycopersicum* genome build SL 4.0 [37] and annotation version ITAG 4.1 available through Sol Genomics Network (<https://solgenomics.net/>) [38]. Sequencing and mapping statistics for all samples are provided in Supplementary Table 1. STAR categorizes some reads as 'unmapped: too short', which indicates that, after trimming and alignment, the remaining read length was insufficient for reliable mapping (below the seed length threshold). These fractions were low across samples (generally < 5%) and are typical of Illumina RNA-seq datasets. Reads mapping to multiple loci were retained by STAR's default settings but were excluded from downstream DEG analyses, which relied on uniquely mapped reads. All downstream analyses were conducted using quality- and length-filtered reads that mapped uniquely to the tomato reference genome. Aligned reads were counted and assigned to gene meta-features using the program RSEM [39]. Count files were imported into the R programming environment and analyzed for differentially expressed genes using edgeR [40]. Low-expressed genes were removed using the filterByExprs() function from edgeR, which accounts for group-wise library sizes and retains genes with adequate expression in a minimum number of samples, ensuring that downstream statistical analyses focus on robust, biologically meaningful signals. Genes with False Discovery Rates (FDR) ≤ 0.1 and log Fold Change (logFC) $\geq \pm 2$ were identified as differentially expressed genes (DEGs). Venn diagrams comparing lists of DEGs from different contrasts were generated via Venny 2.1 [41]. K-means clustering was applied to differentially expressed genes (DEGs) from specific pairwise contrasts (e.g., Flight vs. Ground, Blue vs. Red) to identify expression trends and functional categories responsive to spaceflight and/or light. This approach allowed us to resolve condition-specific patterns in subsets of genes most responsive to the tested contrasts. K-means clustering plots showing clusters of DEGs were generated using the R package 'cluster' [42]. Gene expression values used for clustering were log-transformed counts per million (logCPM) obtained from the edgeR pipeline. The optimal number of clusters was determined using the Silhouette method [43–45]. Heatmaps depicting patterns of gene expression and clusters across comparisons were generated using the R package pheatmap [46], using log₂-transformed CPM (log₂CPM) values as input. For visualization, expression values were z-score scaled by gene (row) to highlight relative

expression differences across samples. To assess cluster quality, we calculated silhouette score using the silhouette() function from the cluster R package [42]. The silhouette score measures how well each gene fits within its assigned cluster, ranging from -1 (poor fit) to +1 (strong fit). For each cluster, we computed the average silhouette score as a summary metric of cohesion and separation. This value reflects the mean silhouette score of all genes assigned to a given cluster and was used to evaluate clustering consistency across the DEG sets. GO enrichment analysis of K-means clusters derived from DEGs unique and common to various Venn comparisons was performed using the 'enricher' function of R package clusterProfiler (v4.8.2) [46, 47] along with a high-coverage GO functional annotation of tomato protein-coding genes by [48]. GO terms with p.adjust < 0.05 for a minimum of 2 genes were represented using dot plots. KEGG enrichment analysis was performed using tomato 'UniProtKB. TrEMBL.ID' retrieved from plant biomaRt using R package 'biomaRt' [49]. Finally, 'enrichKEGG' function from clusterProfiler (v4.8.2) was used for KEGG enrichment analysis with an adjusted *p*-value < 0.1.

Functional group identification

To assign functional groups to clusters of DEGs, we combined expression-based clustering with annotation and GO enrichment analysis. Genes were grouped into clusters by K-means, with the number of clusters determined using the Silhouette method. Tomato gene IDs were then mapped to the ITAG4.1 annotation (<https://solgenomics.net/>), which provides functional descriptions inferred primarily from sequence similarity and domain prediction. Therefore, we emphasize that functional categories represent predicted rather than experimentally validated roles. To identify dominant biological themes, gene ontology (GO) enrichment analysis was performed using clusterProfiler (v4.8.2), with significance defined as p.adjust < 0.05. Enriched terms were consolidated into broader, biologically interpretable functional groups (e.g., oxidative stress, related terms grouped under "ROS detoxification"). For summary tables, we highlighted functional groups that were most frequent and statistically supported within a cluster. Representative genes were selected from each cluster based on differential expression (FDR ≤ 0.1 & log₂FC ≥ 2), and consistent annotation, serving as illustrative examples of cluster-level functional trends.

Weighted Gene Co-expression Network Analysis (WGCNA)

Weighted Gene Co-expression Network Analysis (WGCNA; v1.7) [51] was used to assess higher-order gene expression patterns across all tissues and conditions. Unlike k-means clustering, which was limited to DEGs from specific pairwise comparisons, WGCNA was

performed on all genes with sufficient variance, independent of DEG status, to construct a co-expression network. This approach enabled the identification of modules representing broader, system-level transcriptional programs that may not be apparent from pairwise DEG clustering alone. Before network construction, we filtered the dataset to remove genes with low variance across treatments and little or no expression. To ensure our network met the scale-free assumptions made by WGCNA, we selected a soft thresholding power of 20 and created an adjacency matrix using this value. This adjacency matrix was converted to a topological overlap matrix, and hierarchical clustering was used to assign genes to modules based on their expression patterns. To simplify downstream analysis, highly similar modules (correlation > 0.7) were merged using hierarchical clustering.

Results

For the results described below abbreviated terms from Table 3 were used frequently, unless otherwise specified, the abbreviated terms always refer to the comparison being made.

RNA sequencing was performed on leaf and adventitious root tissues from tomato plants grown under red-rich and blue-rich lighting, both aboard the ISS and on the ground at KSC. A Principal Components Analysis (PCA) plot comparing tissue type (leaf and adventitious root; PC1) versus condition (flight and ground control; PC2) indicated that the two principal components represented 41.25% of the variation in gene expression across all samples (Fig. 2). The samples separated by tissue type

along PC1, representing approximately 25% of the variance, suggesting tissue-specific gene expression was the primary source of variation in the gene expression data (Fig. 2A). When comparing lighting conditions between flight and ground controls, the blue-rich lighting samples grown in flight showed greater variation compared to those grown on the ground (Fig. 2B). In contrast, samples grown under the red-rich light regime clustered more closely, suggesting that red-rich light may induce a more consistent gene expression response, regardless of condition (Fig. 2C). For completeness, we also performed a global PCA including all samples across tissues, light treatments, and flight conditions (Supplemental Fig. 18). This analysis confirmed that tissue identity is the primary driver of variance, with light and condition effects emerging secondarily, consistent with the stratified PCAs shown in Fig. 2.

We performed differential gene expression analysis between flight and ground controls for each tissue type and compared these differentially expressed genes (DEGs) between light treatments. In the leaves, there were 198 (115 up- and 83 down-regulated) shared DEGs between red-rich and blue-rich light treatments (Fig. 3A and B). In the adventitious roots, we found 305 (261 up- and 44 down-regulated) shared DEGs between light treatments (Fig. 3C and D). Additionally, regardless of tissue type or light treatment, we consistently observed more upregulated genes in the flight samples for all comparisons (Fig. 3B and D). We also performed differential gene expression analysis between red-rich and blue-rich light treatments for each tissue type and compared these

Table 3 Abbreviated terms used for showing different comparisons and their corresponding detailed contrasts as used in the differential gene expression analysis pipeline, where red and blue refer to plants grown under red-rich or blue-rich lighting treatments

Abbreviated Term	Samples Represented by the Abbreviated Terms	Underlying Comparison	Detailed Comparisons (Treatment vs. Control)
LeafBlue	DEGs identified in leaves from spaceflight-grown plants under blue-rich light	Flight vs. Ground	Flight.Leaf.Blue vs. Ground.Leaf.Blue
LeafRed	DEGs identified in leaves from spaceflight-grown plants under red-rich light	Flight vs. Ground	Flight.Leaf.Red vs. Ground.Leaf.Red
LeafBlue_LeafRed	DEGs common to LeafBlue and LeafRed	Flight vs. Ground	
Adv.RootBlue	DEGs identified in adventitious roots from spaceflight-grown plants under blue-rich light	Flight vs. Ground	Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue
Adv.RootRed	DEGs identified in adventitious roots from spaceflight-grown plants under red-rich light	Flight vs. Ground	Flight.Adv.Root.Red vs. Ground.Adv.Root.Red
Adv.RootBlue_Adv.RootRed	DEGs common to Adv.RootBlue and Adv.RootRed	Flight vs. Ground	
FltLeaf	DEGs identified in leaves from spaceflight-grown plants comparing blue-rich vs. red-rich light	Blue vs. Red	Flight.Leaf.Blue vs. Flight.Leaf.Red
GndLeaf	DEGs identified in leaves from ground-grown plants comparing blue-rich vs. red-rich light	Blue vs. Red	Ground.Leaf.Blue vs. Ground.Leaf.Red
FltLeaf_GndLeaf	DEGs common to FltLeaf and GndLeaf	Blue vs. Red	
FltAdv.Root	DEGs identified in adventitious roots from spaceflight-grown plants comparing blue-rich vs. red-rich light	Blue vs. Red	Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red
GndAdv.Root	DEGs identified in adventitious roots from ground-grown plants comparing blue-rich vs. red-rich light	Blue vs. Red	Ground.Adv.Root.Blue vs. Ground.Adv.Root.Red
FltAdv.Root_GndAdv.Root	DEGs common to FltAdv.Root and GndAdv.Root	Blue vs. Red	

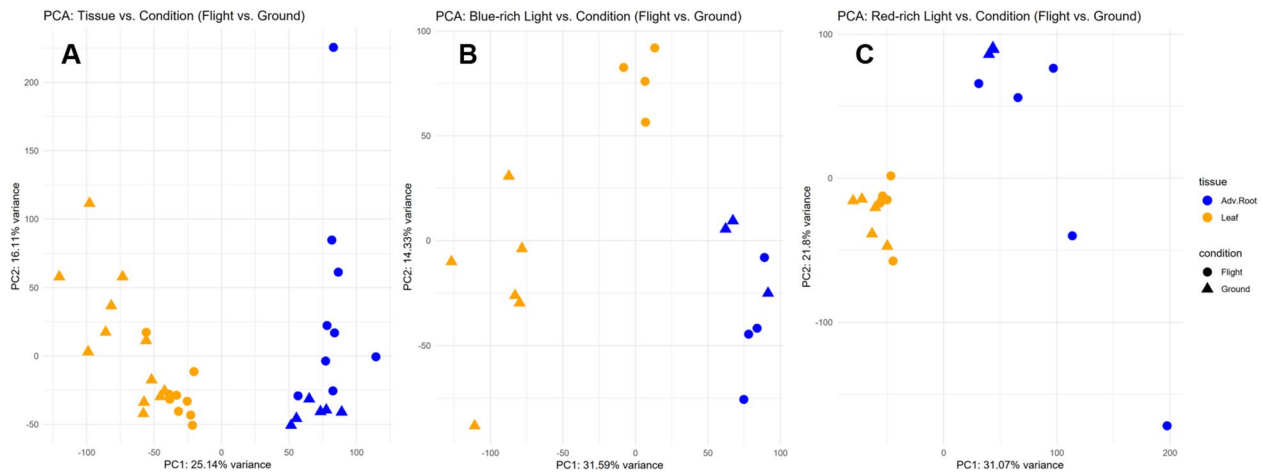


Fig. 2 Principal component analysis (PCA) of RNA-seq samples. PCA plots showing the separation of RNA-seq samples by tissue type and experimental condition. **A** PCA of all samples comparing leaf and adventitious root tissues under flight and ground conditions. **B** PCA of samples grown under blue-rich light, showing the effect of flight versus ground conditions. **C** PCA of samples grown under red-rich light, showing the effect of flight versus ground conditions

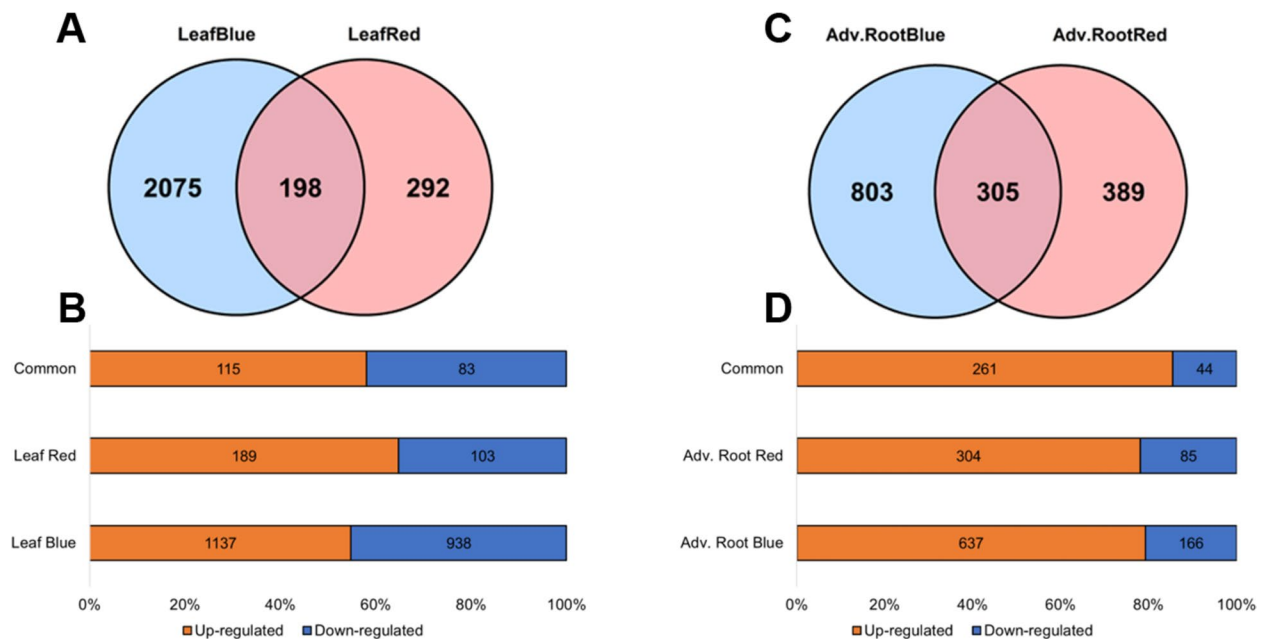


Fig. 3 Venn diagrams and DEG distribution for leaf and adventitious root samples grown under blue-rich or red-rich lighting from flight or ground controls. **A** Venn diagram showing unique and common DEGs for leaf samples. **B** Breakdown of DEGs into up/downregulated categories. **C** Venn diagram for adventitious root samples. **D** Breakdown of DEGs into up/downregulated categories. DEGs were identified with $\log_{2}FC \geq 2$ and $FDR \leq 0.1$

DEGs between locations, i.e. spaceflight vs. ground controls. In the leaves, we identified 45 (22 up- and 23 down-regulated) shared DEGs (Fig. 4A) and in the adventitious roots, we found 105 (24 up- and 81 down-regulated) shared DEGs between flight and ground conditions (Fig. 4C). Furthermore, in the leaves, we observed more DEGs between light treatments on the ground than in flight, however the opposite pattern was observed in the adventitious roots (Fig. 4B and D).

Transcriptional responses to spaceflight in tomato leaves under blue-rich and red-rich lighting

K-means clustering was conducted on differentially expressed genes (DEGs) identified through comparisons between flight and ground control samples under blue-rich and red-rich lighting conditions, across both leaf and adventitious root tissues. DEGs from Venn comparisons were categorized into clusters based on shared expression patterns. For interpretation, a) the results of K-means clustering (.CSV files with genes assigned to specific Silhouette clusters of gene expression), b) PCA

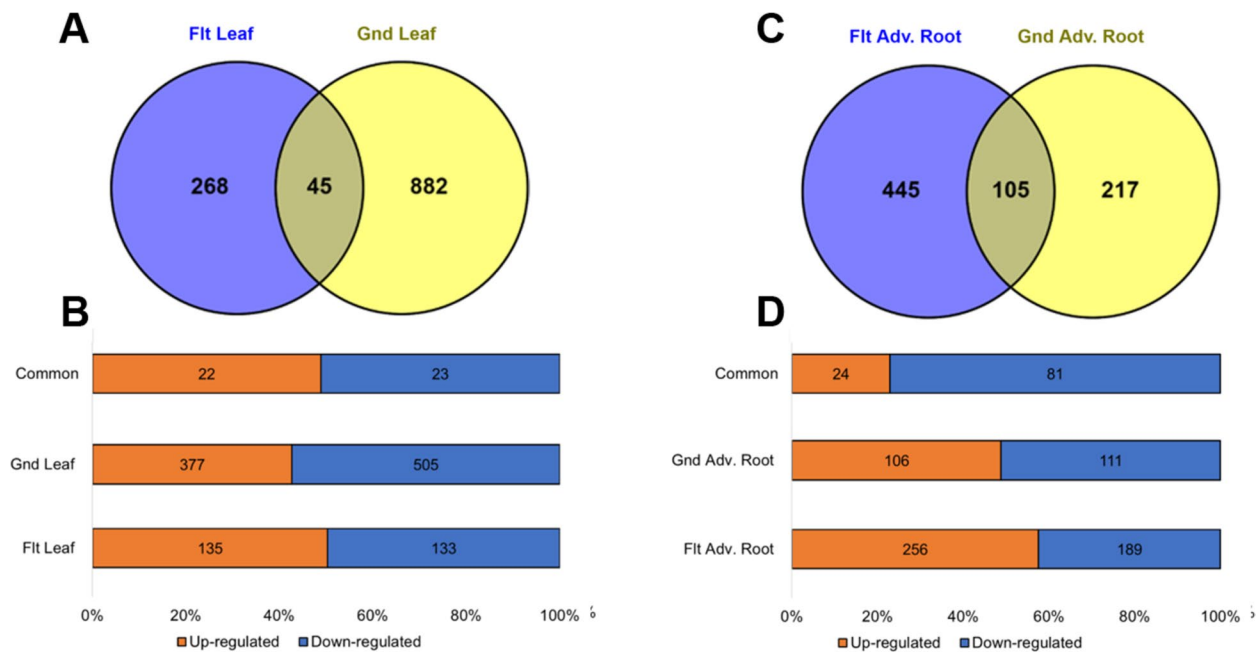


Fig. 4 Venn diagrams and DEG distribution for leaf and adventitious root samples grown either in flight or as ground controls under blue-rich or red-rich lighting. **A** Venn diagram showing unique and common DEGs for leaf samples. **B** Breakdown of DEGs into up/downregulated categories. **C** Venn diagram for adventitious root samples. **D** Breakdown of DEGs into up/downregulated categories. DEGs were identified with $\log_2FC \geq \pm 2$ and $FDR \leq 0.1$

plots showing different clusters, and c) heatmaps detailing expression patterns of DEGs from each cluster were integrated together to explain distinct clustering patterns across the blue-rich and red-rich lighting conditions for both leaf and adventitious root samples (Figs. 5 and 6, Table 4, Supplemental Figs. 1-2).

In LeafBlue, four clusters revealed distinct gene expression dynamics. Genes in cluster 1 were upregulated in flight sample exhibiting stable expression linked to fundamental cellular processes (e.g., cell wall proteins, ribosomal proteins). Cluster 2 genes were upregulated in ground controls involving stress-responsive proteins and defense enzymes identified based on similarity of functional groups. Cluster 3 displayed high variability, indicating condition-specific regulation (e.g., hormonal signaling regulators, transcription factors). Cluster 4 showed low expression, representing conditionally inactive pathways (e.g., photosynthesis-related proteins) (Fig. 5A; Supplemental Fig. 1).

In LeafRed samples, cluster 1 maintained baseline metabolic stability (e.g., cytochrome P450 enzymes), while cluster 2 highlighted flight-induced upregulation of stress-related genes (e.g., bHLH transcription factors). Cluster 3 was highly variable, associated with developmental regulation and secondary metabolite synthesis. Cluster 4 featured genes with low expression, linked to nutrient transport and energy metabolism (Fig. 5B; Supplemental Fig. 2). Comparative analyses revealed that LeafBlue emphasized stress response and environmental adaptation, whereas LeafRed prioritized developmental

regulation and hormonal pathways. Together, these results provide a nuanced understanding of transcriptional regulation across environmental and physiological contexts, balancing metabolic stability, adaptation, and development.

Spectral light effects on gene expression in adventitious roots

For adventitious root samples, the blue-rich condition showed separation into four clusters. This clustering further reinforced the impact of blue-rich lighting on gene expression profiles, regardless of tissue type (Fig. 5C; Supplemental Fig. 3). In the red-rich adventitious root samples, while the clusters were more distinct than those seen in the leaf counterparts, the separation remains less striking compared to the blue-rich condition (Fig. 5D; Supplemental Fig. 4).

The Adv.RootBlue samples showed four distinct clusters of gene expression dynamics. Cluster 1 genes were mainly upregulated in flight samples, suggesting their involvement in stress adaptation to microgravity. Functional categories included pathogenesis-related proteins, RING-finger ubiquitin ligases, and chlorophyll a/b-binding proteins. Cluster 2 genes exhibited stable expression across conditions, likely representing essential cellular processes, such as ATP synthase subunits, ribosomal proteins, and histone-related proteins. Cluster 3 genes displayed variable expression between flight and ground samples, highlighting condition-specific regulation. Functional groups included cytochrome P450 enzymes,

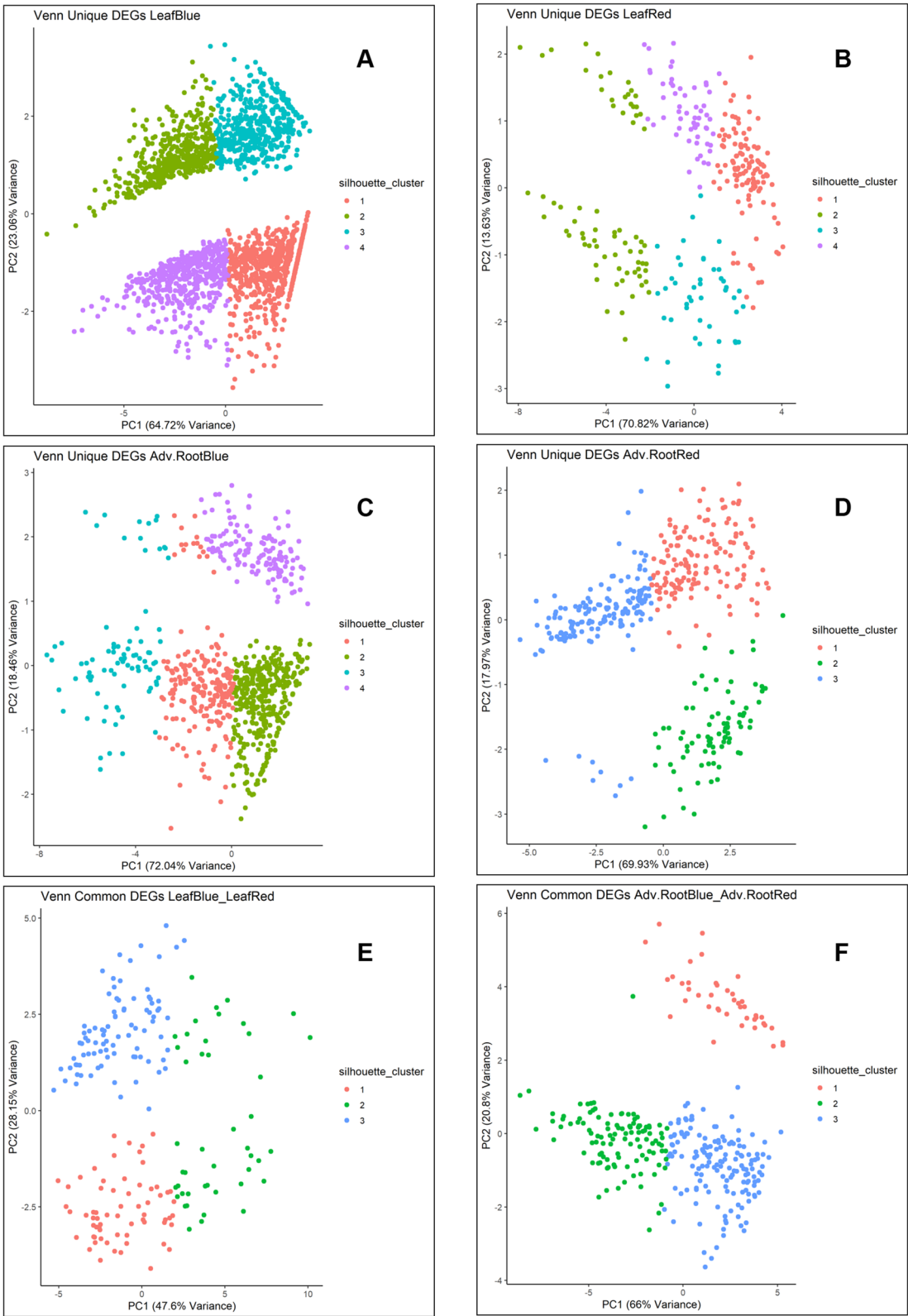


Fig. 5 (See legend on next page.)

(See figure on previous page.)

Fig. 5 Principal component analysis (PCA) plots showing k-means clustering of differentially expressed genes (DEGs). Each point represents a gene, colored by its assigned expression cluster. Panels show DEGs identified from: **(A)** Flight.Leaf.Blue vs. Ground.Leaf.Blue comparison, **(B)** Flight.Leaf.Red vs. Ground.Leaf.Red comparison, **(C)** Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue comparison, **(D)** Flight.Adv.Root.Red vs. Ground.Adv.Root.Red comparison, **(E)** DEGs common to both Flight.Leaf.Blue vs. Ground.Leaf.Blue and Flight.Leaf.Red vs. Ground.Leaf.Red comparisons, **(F)** DEGs common to both Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue and Flight.Adv.Root.Red vs. Ground.Adv.Root.Red comparisons. The optimal number of clusters was determined using the Silhouette method

glycosyltransferases, and phenylalanine ammonia-lyases. Cluster 4 genes were predominantly upregulated in ground samples, featuring auxin response factors, zinc finger transcription factors, and DNA repair enzymes (Fig. 5C; Table 5; Supplemental Fig. 3).

In Adv.RootRed samples, three clusters of gene expression were identified. Cluster 1 genes showed moderate upregulation in flight samples, likely linked to adaptation to microgravity-induced stress. Functional groups included lipid metabolism, stress-response enzymes, and secondary metabolite pathways. Cluster 2 genes were markedly upregulated in flight samples and minimally expressed in ground conditions, encompassing antioxidant pathways, phytohormone signaling, and energy metabolism enzymes. Cluster 3 genes exhibited high variability, with consistent upregulation in flight samples and reduced expression in ground samples. Functional categories included cell wall biosynthesis, ROS detoxification, and nutrient transport systems. These findings demonstrate the transcriptional response to lighting conditions and flight, emphasizing stress adaptation, metabolic stability, and condition-specific regulatory mechanisms (Fig. 5D, Table 5, Supplemental Fig. 4).

Conserved blue-rich vs. red-rich light-dependent gene expression patterns in leaf and adventitious root

To simplify references, we use the terms LeafBlue_LeafRed and Adv.RootBlue_Adv.RootRed to refer to DEGs that are common to both blue-rich and red-rich light conditions in flight vs. ground comparisons specifically, the overlap of “Flight.Leaf.Blue vs. Ground.Leaf.Blue” and “Flight.Leaf.Red vs. Ground.Leaf.Red” for leaves, and of “Flight.Adv.Root.Blue vs. Ground. Adv.Root.Blue” and “Flight. Adv.Root.Red vs. Ground. Adv.Root.Red” for adventitious roots (Table 3). For the 198 and 305 DEGs shared across these conditions in leaves and adventitious roots, respectively (from the Venn intersections in Fig. 3A and C), PCA plots and heatmaps revealed three distinct gene expression clusters (Fig. 5E–F; Supplemental Figs. 5 and 6). For LeafBlue_LeafRed comparison, cluster 1 genes exhibited balanced expression between LeafBlue and LeafRed. Functional categories included stress response, ATP synthesis, and ribosomal protein activity. This cluster represents common pathways essential for both lighting conditions. Cluster 2 genes showed stronger expression in LeafBlue. Functional groups

encompassed enhanced photosynthetic efficiency, ABA signaling, and secondary metabolism, highlighting LeafBlue’s adaptability to environmental fluctuations. Cluster 3 genes had higher expression in LeafRed, linked to pathogen resistance, cell wall remodeling, and auxin-mediated growth, emphasizing LeafRed’s role in structural integrity and biotic stress defense (Fig. 5E; Table 6; Supplemental Fig. 5).

In the comparison between Adv.RootBlue and Adv.RootRed, cluster 1 genes were mainly differentially expressed in Adv.RootBlue. These genes were associated with functional groups such as oxidative stress response, protein degradation, and stress signaling, highlighting Adv.RootBlue’s emphasis on stress tolerance. Cluster 2 genes showed balanced expression across both conditions. Functional categories included secondary metabolite synthesis, antimicrobial activity, and ABA-mediated signaling, supporting general stress adaptation. Cluster 3 genes were found to be upregulated in Adv.RootRed represented by functional groups such as cell wall remodeling, lipid metabolism, and pathogen resistance, emphasizing Adv.RootRed’s role in structural fortification and immune readiness (Fig. 5F; Table 7; Supplemental Fig. 6).

Distinct light-responsive gene expression clusters in leaf tissues under flight and ground conditions

Like Flight vs. Ground comparisons discussed above, DEGs unique and common to each of the Venn comparisons from Fig. 4A and C were divided into clusters and analyzed for patterns of gene expression (Fig. 6; Supplemental Figs. 7–12). DEGs from FltLeaf and GndLeaf samples under blue- and red-rich lighting conditions revealed distinct transcriptional patterns divided into four clusters via PCA and heatmap analyses. For FltLeaf, cluster 1 showed elevated expression in red-rich lighting condition, involving auxin-responsive signaling and cell wall remodeling, suggesting auxin-mediated growth and stress adaptation. Cluster 2 highlighted enhanced expression in blue-rich lighting condition, linked to amino acid transport and secondary metabolite biosynthesis, indicating metabolic reprogramming for growth. Cluster 3 exhibited somewhat balanced expression across lighting conditions, with genes involved in transcriptional regulation and oxidative stress management. Cluster 4 showed high expression in red-rich lighting condition, associated with stress signaling and ATP generation (Fig. 6A; Table 8; Supplemental Fig. 7).

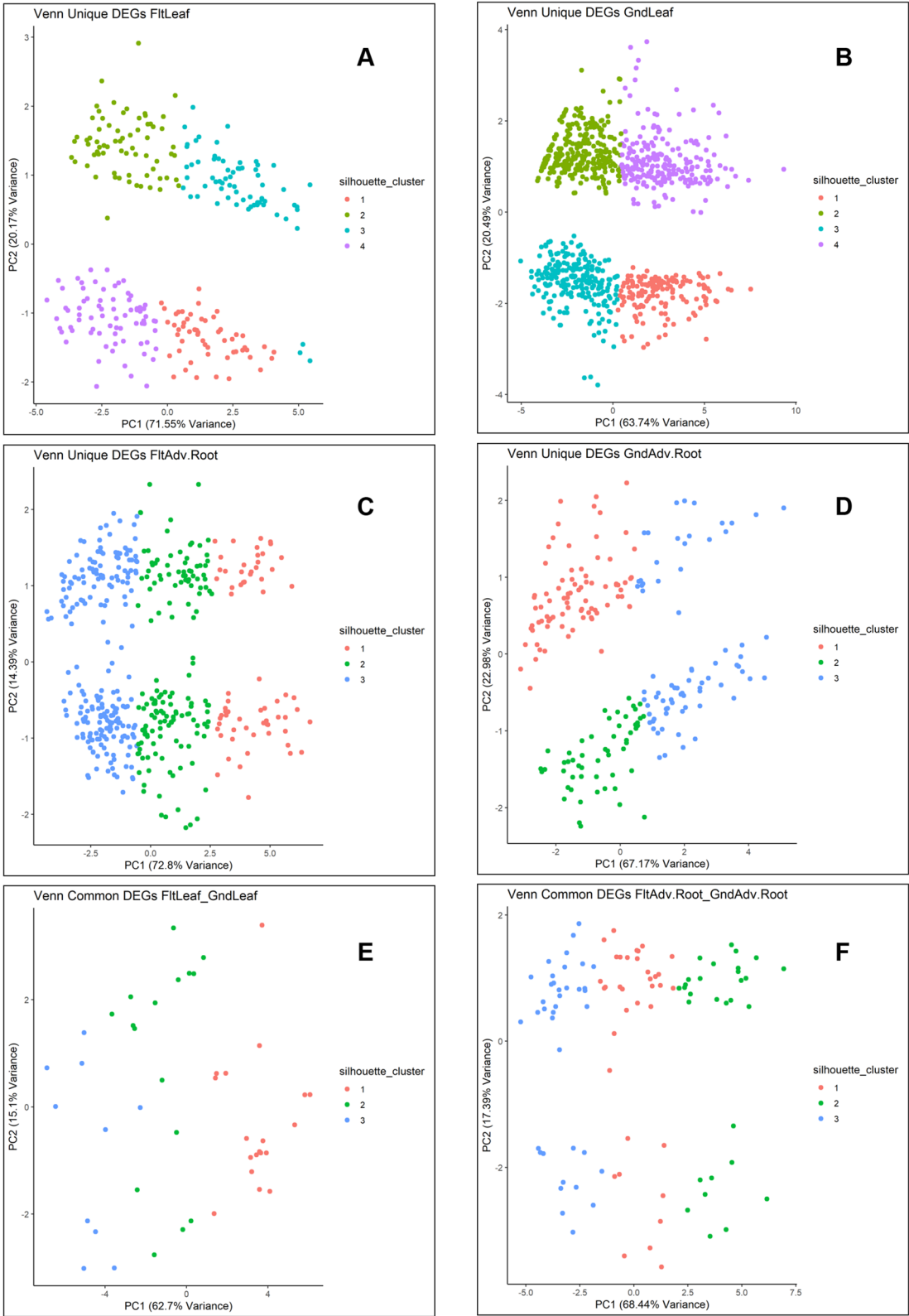


Fig. 6 (See legend on next page.)

(See figure on previous page.)

Fig. 6 Principal component analysis (PCA) plots showing k-means clustering of differentially expressed genes (DEGs). Each point represents a gene, colored by its assigned cluster based on expression profiles. Panels show DEGs from: (A) Flight.Leaf.Blue vs. Flight.Leaf.Red comparison, (B) Ground.Leaf.Blue vs. Ground.Leaf.Red comparison, (C) Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red comparison, (D) Ground.Adv.Root.Blue vs. Ground.Adv.Root.Red comparison, (E) DEGs shared between Flight.Leaf.Blue vs. Flight.Leaf.Red and Ground.Leaf.Blue vs. Ground.Leaf.Red comparisons, (F) DEGs shared between Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red and Ground.Adv.Root.Blue vs. Ground.Adv.Root.Red comparisons. The optimal number of clusters was determined using the Silhouette method

For GndLeaf, cluster 1 was enriched in blue-rich light, featuring genes involved in hormone signaling and photosynthesis, reflecting enhanced photosynthetic capacity and stress adaptation. Cluster 2 showed consistent high expression under red-rich lighting, linked to metabolic maintenance pathways like secondary metabolite biosynthesis and protein turnover. Cluster 3 displayed higher expression in blue-rich lighting, involving pathogen defense and phenylpropanoid biosynthesis. Cluster 4 was enriched in red-rich condition, linked to structural reorganization and auxin signaling. These findings underscore blue-rich lighting's role in growth and metabolic activity and red-rich lighting's focus on structural integrity and stress responses, highlighting condition-specific transcriptional dynamics in leaf tissues (Fig. 6B; Table 8; Supplemental Fig. 8).

Stress and structural gene expression programs induced by light quality in flight and ground adventitious roots

The Silhouette clustering of gene expression in FltAdv.Root and GndAdv.Root datasets revealed three distinct clusters, each characterized by unique functional categories and genes. In the FltAdv.Root dataset, cluster 1 showed elevated expression of genes related to stress responses and metabolic reprogramming. Cluster 2 was enriched for genes involved in biosynthesis and modification of structural cell wall polysaccharides, including cellulose and cutin, suggesting reinforcement of structural integrity in spaceflight conditions. Cluster 3 exhibited balanced expression, reflecting roles in metabolic regulation and detoxification (Fig. 6C; Table 9; Supplemental Fig. 9). Similarly, in the GndAdv.Root dataset, Cluster 1 was enriched in genes related to oxidative stress detoxification (e.g., peroxidases, oxidoreductases). Cluster 2 was characterized by metabolic processes including amino acid biosynthesis and carbohydrate metabolism. Cluster 3 was associated with structural remodeling through cell wall organization and lignin biosynthesis, coupled with redox homeostasis. These findings underscore the interplay between oxidative stress adaptation, metabolic reprogramming, and structural adjustments in root responses to environmental challenges (Fig. 6D; Table 9; Supplemental Fig. 10).

Clustering analysis of common flight-regulated DEGs in leaf and adventitious root samples

Similarly, the terms FltLeaf_GndLeaf and FltAdv.Root_GndAdv.Root refer to DEGs common to blue-rich vs. red-rich comparisons under flight and ground conditions, respectively (i.e., the intersection of “Flight.Leaf.Blue vs. Flight.Leaf.Red” and “Ground.Leaf.Blue vs. Ground.Leaf.Red” for leaves, and “Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red” and “Ground.Adv.Root.Blue vs. Ground.Adv.Root.Red” similarly for adventitious roots). The clustering analysis of DEGs common to flight and ground control conditions revealed distinct transcriptional and functional profiles in both FltLeaf_GndLeaf and FltAdv.Root_GndAdv.Root datasets (Fig. 6E–F; Supplemental Figs. 11–12). For the FltLeaf_GndLeaf comparison, PCA and K-means clustering of transcriptomic data identified three distinct clusters, revealing functional divergence between FltLeaf and GndLeaf conditions. Cluster 1 was enriched for genes involved in oxidative stress responses (e.g., ascorbate oxidase) and biosynthetic pathways (e.g., Caffeoyl-CoA O-methyltransferase), showing higher expression in FltLeaf. Cluster 2 comprised genes with intermediate expression levels across Flight and Ground leaves (e.g., chitinases, citrate-binding proteins). These genes did not show strong directional changes but instead indicated partial overlap in metabolic and stress-related pathways between conditions. Cluster 3 genes were downregulated in GndLeaf (e.g., ethylene response factor C.6), suggesting suppressed pathways unique to FltLeaf (Fig. 6E; Table 10; Supplemental Fig. 11).

For the FltAdv.Root_GndAdv.Root comparison three distinct clusters were identified. Cluster 1 genes with high expression were linked to stress adaptation and biosynthetic pathways, including 1-aminocyclopropane-1-carboxylate synthase 3 and amino acid transporter. These genes are upregulated in FltAdv.Root compared to GndAdv.Root. Cluster 2 genes were found to be involved in energy metabolism and cellular signaling, such as Agamous-like MADS-box protein AGL1, exhibit moderate expression across conditions, suggesting overlap in functional pathways. Cluster 3 genes were found to be downregulated in GndAdv.Root, such as actin cross-linking protein and ATPase subunit 4. This cluster may represent pathways specific to growth under FltAdv.Root conditions (Fig. 6F; Table 10; Supplemental Fig. 12).

Table 4 Summary of gene expression clusters in LeafBlue and LeafRed samples under various conditions. Clusters are categorized based on expression stability, functional highlights, associated functional groups, and representative gene IDs. Functional groups were derived from ITAG4.1 annotations and supported by GO enrichment analyses (adjusted $p < 0.05$), with redundant terms condensed into broader categories for clarity. Representative genes were selected based on differential expression ($FDR \leq 0.1$ & $|\log_2 FC| \geq 2$) and annotation clarity and are presented as examples of each cluster's inferred biology rather than exhaustive markers

Cluster #	Number of Genes	Avg. Silhouette Score	Cluster Highlights	Functional Groups	Representative Gene IDs
LeafBlue; Fig. 5A and Suppl. Figure 1A-D					
1	484	0.3342	Upregulated in flight samples	Cell wall structural proteins Metabolic enzymes Ribosomal proteins	Solyc08g081535.1; Solyc01g058250.2; Solyc02g062490.4
2	476	0.4108	Variably expressed; upregulated in ground controls	Stress-responsive proteins Defense-related enzymes	Solyc04g071590.3; Solyc01g095030.3; Solyc11g066950.2
3	462	0.3023	Highly variable; tightly regulated, condition-specific processes	Regulators of secondary metabolite pathways Hormonal signaling regulators Growth-related proteins Transcription factors	Solyc02g083520.3; Solyc03g097820.2; Solyc04g007300.4
4	653	0.3653	Low expression; conditional inactivity	Nutrient storage proteins Photosynthesis-related proteins Transport proteins	Solyc05g014390.4; Solyc07g065320.4; Solyc07g066010.3
LeafRed; Fig. 5B and Suppl. Figure 2A-D					
1	62	0.2698	Stable expression; baseline metabolic processes	Cytochrome P450 enzymes Ribosomal protein S4 Histone proteins	Solyc01g109150.4; Solyc02g038690.1; Solyc03g121100.4
2	107	0.3637	Variable expression; upregulated under flight conditions	bHLH transcription factor 107 Hydroxyproline-rich glycoproteins ATP-dependent DNA helicase	Solyc10g081980.2; Solyc10g081970.3; Solyc12g010960.2
3	72	0.3322	Highly variable; developmental regulation and secondary metabolite synthesis	Phenylalanine ammonia-lyase Cytochrome P450 enzymes Zinc finger transcription factors	Solyc00g500095.1; Solyc10g161270.1; Solyc10g078770.2
4	51	0.2047	Low expression	Sugar facilitator proteins NADH-ubiquinone oxidoreductase Histone H3 proteins	Solyc09g074300.1; Solyc12g056540.1; Solyc01g086800.4

Table 5 Summary of gene expression clusters in Adv.RootBlue and Adv.RootRed samples under various conditions. Clusters are categorized based on expression stability, functional highlights, associated functional groups, and representative gene IDs. Functional groups were derived from ITAG4.1 annotations and supported by GO enrichment analyses (adjusted $p < 0.05$), with redundant terms condensed into broader categories for clarity. Representative genes were selected based on differential expression ($FDR \leq 0.1$ & $|\log_2 FC| \geq 2$) and annotation clarity, and are presented as examples of each cluster's inferred biology rather than exhaustive markers

Cluster #	Number of Genes	Avg. Silhouette Score	Cluster Description	Functional Groups	Representative Gene IDs
Adv.RootBlue; Fig. 5C and Suppl. Figure 3A-D					
1	234	0.2968	Upregulated in flight samples; stress adaptation	Pathogenesis-related proteins, RING-finger ubiquitin ligases, chlorophyll a/b-binding proteins	Solyc01g097280.3; Solyc02g071010.1; Solyc08g074260.3
2	347	0.3314	Stable expression across flight and ground conditions	ATP synthase subunits, ribosomal proteins, histone-related proteins	Solyc00g500050.1; Solyc00g500133.1; Solyc00g500203.1
3	138	0.3250	Variably expression between flight and ground samples	Cytochrome P450 enzymes, glycosyltransferases, phenylalanine ammonia-lyases	Solyc04g071780.3; Solyc05g056170.3; Solyc01g106620.2
4	84	0.4295	Upregulated in ground samples	Auxin response factors, zinc finger transcription factors, DNA repair enzymes	Solyc02g087970.1; Solyc07g053750.1; Solyc10g005470.3
Adv.RootRed; Fig. 5D and Suppl. Figure 4A-C					
1	149	0.2996	Moderate upregulation in flight samples	Lipid metabolism, stress-response enzymes (oxidases and reductases), secondary metabolite pathways	Solyc04g007800.5; Solyc10g076210.2; Solyc01g101050.3
2	160	0.4718	Markedly upregulated in flight samples	Antioxidant pathways, phytohormone signaling, energy metabolism enzymes	Solyc06g072640.1; Solyc04g011690.5; Solyc04g016190.1
3	80	0.3587	High variability; upregulated in flight samples	Cell wall biosynthesis and remodeling enzymes, ROS-scavenging proteins, nutrient transport systems	Solyc09g097960.3; Solyc09g011520.3; Solyc04g071590.3

Table 6 Summary of gene expression clusters in LeafBlue_LeafRed samples under various conditions. Clusters are categorized based on expression stability, functional highlights, associated functional groups, and representative gene IDs. Functional groups were derived from ITAG4.1 annotations and supported by GO enrichment analyses (adjusted $p < 0.05$), with redundant terms condensed into broader categories for clarity. Representative genes were selected based on differential expression ($FDR \leq 0.1$ & $|\log_2FC| \geq 2$) and annotation clarity, and are presented as examples of each cluster's inferred biology rather than exhaustive markers

Cluster #	Number of Genes	Avg. Silhouette Score	Cluster Description	Functional Categories	Representative Gene IDs
LeafBlue_LeafRed; Fig. 5E and Suppl. Figure 5A-C					
1	64	0.2864	Balanced expression between LeafBlue and LeafRed	Stress response, ATP synthesis, ribosomal protein activity	Solyc01g091705.2; Solyc00g021640.2; Solyc05g015800.3
2	89	0.2985	Stronger expression in LeafBlue	Enhanced photosynthetic efficiency, ABA signaling, secondary metabolism, transcriptional regulation, environmental stress responses	Solyc03g093540.1; Solyc02g084850.3; Solyc08g076970.3
3	45	0.2738	Higher expression in LeafRed	Pathogen resistance, cell wall remodeling, auxin-mediated growth	Solyc08g080670.1; Solyc08g080660.1; Solyc12g098225.1

Table 7 Summary of gene expression clusters in Adv.RootBlue_Adv.RootRed samples under various conditions. Clusters are categorized based on expression stability, functional highlights, associated functional groups, and representative gene IDs. Functional groups were derived from ITAG4.1 annotations and supported by GO enrichment analyses (adjusted $p < 0.05$), with redundant terms condensed into broader categories for clarity. Representative genes were selected based on differential expression ($FDR \leq 0.1$ & $|\log_2FC| \geq 2$) and annotation clarity, and are presented as examples of each cluster's inferred biology rather than exhaustive markers

Cluster #	Number of Genes	Avg. Silhouette Score	Cluster Description	Functional Categories	Representative Gene IDs
Adv.RootBlue_Adv.RootRed; Fig. 5F and Suppl. Figure 6A-C					
1	42	0.4105	Predominantly expressed in Adv.RootBlue	Oxidative stress responses, protein degradation, stress signaling, antioxidant activity, ubiquitin-mediated protein turnover	Solyc01g109470.3; Solyc09g090500.3; Solyc10g052712.1
2	112	0.4799	Balanced expression across conditions	Secondary metabolite synthesis, antimicrobial activity, ABA-mediated signaling	Solyc11g066800.2; Solyc07g009510.1; Solyc09g006005.1
3	151	0.3645	Upregulated in Adv.RootRed	Cell wall remodeling, lipid metabolism, pathogen resistance, jasmonic acid-mediated signaling, biotic stress responses	Solyc08g079190.2; Solyc08g079230.1; Solyc03g091000.1

Expression variability across light treatments

The observed transcriptomic variability under blue-rich lighting is particularly noteworthy. Since DEG counts alone could reflect either biological stabilization under red-rich light or a masking effect due to prior red-light activation, we conducted multiple complementary analyses to resolve this ambiguity. First, PCA plot of logCPM values (Supplemental Fig. 13A) revealed greater dispersion among blue-rich light samples, particularly between ground and flight, whereas red-rich light samples showed tighter clustering indicating lower inter-condition variability. Second, gene-level variance distributions supported this pattern, with blue-light samples exhibiting higher median variance (Supplemental Fig. 13B). Third, a generalized linear model with a Light $\times$ Condition interaction term identified genes differentially modulated by flight under distinct lighting, and expression box-plots (Supplemental Fig. 14) confirmed dynamic gene responses under blue-rich light but not red-rich. Fourth, overlap analyses among lighting and flight comparisons

(Supplemental Figs. 15–16) revealed limited gene overlap, suggesting red-rich light does not simply pre-activate flight-responsive genes. Finally, the magnitude of transcriptional change, quantified by mean and cumulative $|\log_2FC|$ (Supplemental Fig. 17), was highest under blue-rich lighting. Collectively, these findings strengthen our conclusion that red-rich light stabilizes, while blue-rich light amplifies, transcriptional responses to space-flight. Blue-rich light is perceived by photoreceptors such as cryptochromes (CRY1, CRY2) and phototropins (PHOT1, PHOT2), which coordinate photomorphogenesis, circadian rhythm regulation, and stress signaling. In microgravity, disruptions in gravitational cues may interfere with the localization or downstream signaling of these receptors. While our dataset did not show strong differential expression of CRY or PHOT transcripts, multiple downstream targets, including HY5 homologs and flavonoid biosynthetic genes, exhibited altered regulation under flight conditions. This suggests that blue light pathways may be transcriptionally or epigenetically

Table 8 Summary of gene expression clusters in FltLeaf and GndLeaf samples under various conditions. Clusters are categorized based on expression stability, functional highlights, associated functional groups, and representative gene IDs. Functional groups were derived from ITAG4.1 annotations and supported by GO enrichment analyses (adjusted $p < 0.05$), with redundant terms condensed into broader categories for clarity. Representative genes were selected based on differential expression ($FDR \leq 0.1$ & $|\log_2 FC| \geq 2$) and annotation clarity, and are presented as examples of each cluster's inferred biology rather than exhaustive markers

Cluster #	Number of Genes	Avg. Silhouette Score	Cluster Description	Functional Categories	Representative Gene IDs
FltLeaf; Fig. 6A and Suppl. Figure 7A-D					
1	58	0.3704	Higher expression in red-rich	Auxin-responsive signaling, glycoproteins involved in cell wall remodeling, hormonal signaling, structural pathways	Solyc12g096570.1; Solyc01g111260.2; Solyc12g010960.2
2	75	0.3142	Higher expression in blue-rich	Amino acid transport, secondary metabolite biosynthesis, metabolic transport, biosynthesis pathways	Solyc11g007890.2; Solyc09g015380.1; Solyc11g069680.1
3	69	0.3703	Balanced expression across both lighting conditions	Transcriptional regulation, oxidative stress management	Solyc03g093550.1; Solyc08g078190.3; Solyc07g055710.3
4	66	0.4021	High expression in red-rich	Stress signaling, ATP generation processes	Solyc01g100440.3; Solyc12g044250.2; Solyc01g090170.5
GndLeaf; Fig. 6B and Suppl. Figure 8A-D					
1	233	0.2735	Enriched expression in blue-rich	Hormone signaling, ion transport, photosynthetic and metabolic pathways	Solyc09g089780.3; Solyc11g072110.2; Solyc11g006300.2
2	223	0.2948	Consistent high expression under red-rich	Secondary metabolite biosynthesis, protein turnover	Solyc10g078770.2; Solyc06g068270.3; Solyc12g017470.2
3	144	0.3929	Mostly higher expression under blue-rich	Defense responses, secondary metabolite production	Solyc03g007200.2; Solyc09g091700.4; Solyc10g085190.2
4	282	0.3043	Consistent high expression under red-rich	Cell wall modification, auxin signaling, structural reorganization, hormonal pathways	Solyc06g062920.3; Solyc02g077880.3; Solyc08g008110.3

Table 9 Summary of gene expression clusters in FltAdv.Root and GndAdv.Root samples under various conditions. Clusters are categorized based on expression stability, functional highlights, associated functional groups, and representative gene IDs. Functional groups were derived from ITAG4.1 annotations and supported by GO enrichment analyses (adjusted $p < 0.05$), with redundant terms condensed into broader categories for clarity. Representative genes were selected based on differential expression ($FDR \leq 0.1$ & $|\log_2 FC| \geq 2$) and annotation clarity and are presented as examples of each cluster's inferred biology rather than exhaustive markers

Cluster #	Number of Genes	Avg. Silhouette Score	Cluster Description	Functional Categories	Representative Gene IDs
FltAdv.Root; Fig. 6C and Suppl. Figure 9A-C					
1	215	0.3204	Balanced expression across both lighting conditions	Stress responses, metabolic reprogramming	Solyc02g084850.3; Solyc07g026650.3; Solyc02g038740.4
2	157	0.2262	Balanced expression across both lighting conditions	Biosynthesis, structural integrity	Solyc06g073080.4; Solyc05g014390.4; Solyc02g087580.3
3	73	0.3989	Consistent higher expression under blue-rich	Metabolic regulation, detoxification	Solyc08g079730.2; Solyc10g078770.2; Solyc03g097440.3
GndAdv.Root; Fig. 6D and Suppl. Figure 10A-C					
1	65	0.3242	Higher expression under red-rich	Oxidative stress responses, environmental adaptation	Solyc01g104110.5; Solyc02g088710.4; Solyc06g009280.1
2	73	0.3096	Higher expression under blue-rich	Metabolic regulation	Solyc08g075490.5; Solyc05g025890.3; Solyc09g098380.3
3	79	0.3945	Balanced expression across both lighting conditions	Structural adaptation, redox homeostasis	Solyc02g036350.3; Solyc02g064690.3; Solyc11g027840.2

modulated by the spaceflight environment, contributing to heightened expression variability, as supported by prior studies emphasizing changes in photoreceptor dynamics and downstream effects under similar conditions [50, 51].

GO enrichment analyses

Enrichment of stress, metabolic, and developmental pathways in light- and flight-responsive gene sets

GO enrichment analyses were performed on k-means clusters generated from the unique and common DEGs from flight and ground grown leaf samples under blue-rich or red-rich lighting to get comparative functional

Table 10 Summary of gene expression clusters in FltLeaf_GndLeaf and FltAdv.Root_GndAdv.Root samples under various conditions. Clusters are categorized based on expression stability, functional highlights, associated functional groups, and representative gene IDs. Functional groups were derived from ITAG4.1 annotations and supported by GO enrichment analyses (adjusted $p < 0.05$), with redundant terms condensed into broader categories for clarity. Representative genes were selected based on differential expression ($FDR \leq 0.1$ & $|\log_2FC| \geq 2$) and annotation clarity, and are presented as examples of each cluster's inferred biology rather than exhaustive markers

Cluster #	Number of Genes	Avg. Silhouette Score	Cluster Description	Functional Categories	Representative Gene IDs
FltLeaf_GndLeaf; Fig. 6E and Suppl. Figure 11A-C					
1	19	0.4306	Stress response and biosynthetic pathways	Ascorbate oxidase, Caffeoyl-CoA O-methyltransferase	Solyc04g054690.3; Solyc02g093250.3; Solyc10g074440.2
2	10	0.2226	Moderately expressed defense and signaling pathways	Chitinase, Citrate-binding protein	Solyc01g105650.5; Solyc02g067810.1; Solyc12g008900.2
3	16	0.1818	Downregulated pathways in GndLeaf	Ethylene response factor	Solyc02g079720.1; Solyc04g008370.4; Solyc02g079490.3
FltAdv.Root_GndAdv.Root; Fig. 6F and Suppl. Figure 12A-C					
1	29	0.3844	Stress adaptation and biosynthetic pathways	1-aminocyclopropane-1-carboxylate synthase 3, amino acid transporter	Solyc01g098060.1; Solyc02g091820.3; Solyc02g092450.3
2	37	0.2119	Energy metabolism and cellular signaling	Agamous-like MADS-box protein AGL1	Solyc02g091990.3; Solyc04g077460.3; Solyc03g121540.3
3	39	0.2961	Downregulated pathways in GndAdv. Root	Actin cross-linking protein, ATPase subunit 4	Solyc02g067510.3; Solyc00g500302.1; Solyc00g020040.1

insights into differential effect(s) of flight vs. ground environments.

LeafBlue's four clusters revealed enrichment for processes including spindle elongation, photosynthesis, defense responses, phosphate starvation, secondary metabolism, circadian rhythm, and salicylic acid biosynthesis, with associated molecular functions such as microtubule motor activity, transcription regulation, and NADH dehydrogenase activity (Fig. 7A-D). LeafRed's 292 DEGs distributed into 4 clusters analyzed for GO enrichment identified roles in microtubule-based movement, fruit ripening, terpenoid biosynthesis, sugar transport, cell wall modification, and ethylene signaling, emphasizing developmental regulation and stress responses (Fig. 7E-H). For the 803 DEGs in the Adv. RootBlue dataset, GO-enriched biological processes included the mitotic cell cycle, cellulose catabolism, diterpenoid biosynthesis, oxidative stress response, and ABA signaling, with molecular functions associated with chromatin structure, heme binding, and protein kinase activity (Fig. 8A-D). The three clusters of Adv.RootRed (389 DEGs) showed significant enrichment in processes such as flavanol biosynthesis, pigment production, detoxification, oxidative stress response, and hormonal signaling, with molecular functions including UDP-glycosyltransferase and peroxidase activities (Fig. 8E-G). GO enrichment analysis of DEGs common to LeafBlue and LeafRed (198 DEGs) revealed processes related to transcriptional regulation, ion transport, carbohydrate metabolism, hormonal signaling, and biotic stress resistance, with molecular functions involving oxidoreductase

and protein dimerization activities (Fig. 9A-C). Similarly, DEGs common to Adv.RootBlue and Adv.RootRed (305 DEGs) exhibited shared enrichment in flavonoid biosynthesis, lipid metabolism, oxidative stress response, glutathione metabolism, and hormone signaling, highlighting common adaptive pathways (Fig. 9D-F).

Functional enrichment of light-regulated genes in leaf and adventitious roots under flight and ground conditions

Similar to the Flight vs. Ground comparison done above, GO enrichment analysis was also done on unique and common DEGs associated with blue-rich and/or red-rich light grown leaf and adventitious root samples from flight and/or ground grown plants. In FltLeaf samples, three clusters revealed enrichment in jasmonic acid signaling, oxygen response, defense regulation, secondary metabolite biosynthesis, aromatic compound biosynthesis, oxylipin biosynthesis, fatty acid biosynthesis, and stress resistance, with molecular functions such as monooxygenase activity, redox regulation, protein stability, DNA repair, and iron ion binding (Fig. 10A-C). GndLeaf samples exhibited four clusters enriched in processes such as phosphate starvation response, secondary metabolism, defense mechanisms, cell wall modification, photosynthesis, photoreception, circadian regulation, abiotic stress responses, and flower development, with molecular functions such as redox regulation, phosphate metabolism, chlorophyll binding, and photoreceptor activity (Fig. 10D-G). Flight-grown adventitious roots from blue-rich conditions showed three clusters enriched for biotic stress resistance, structural remodeling, cell

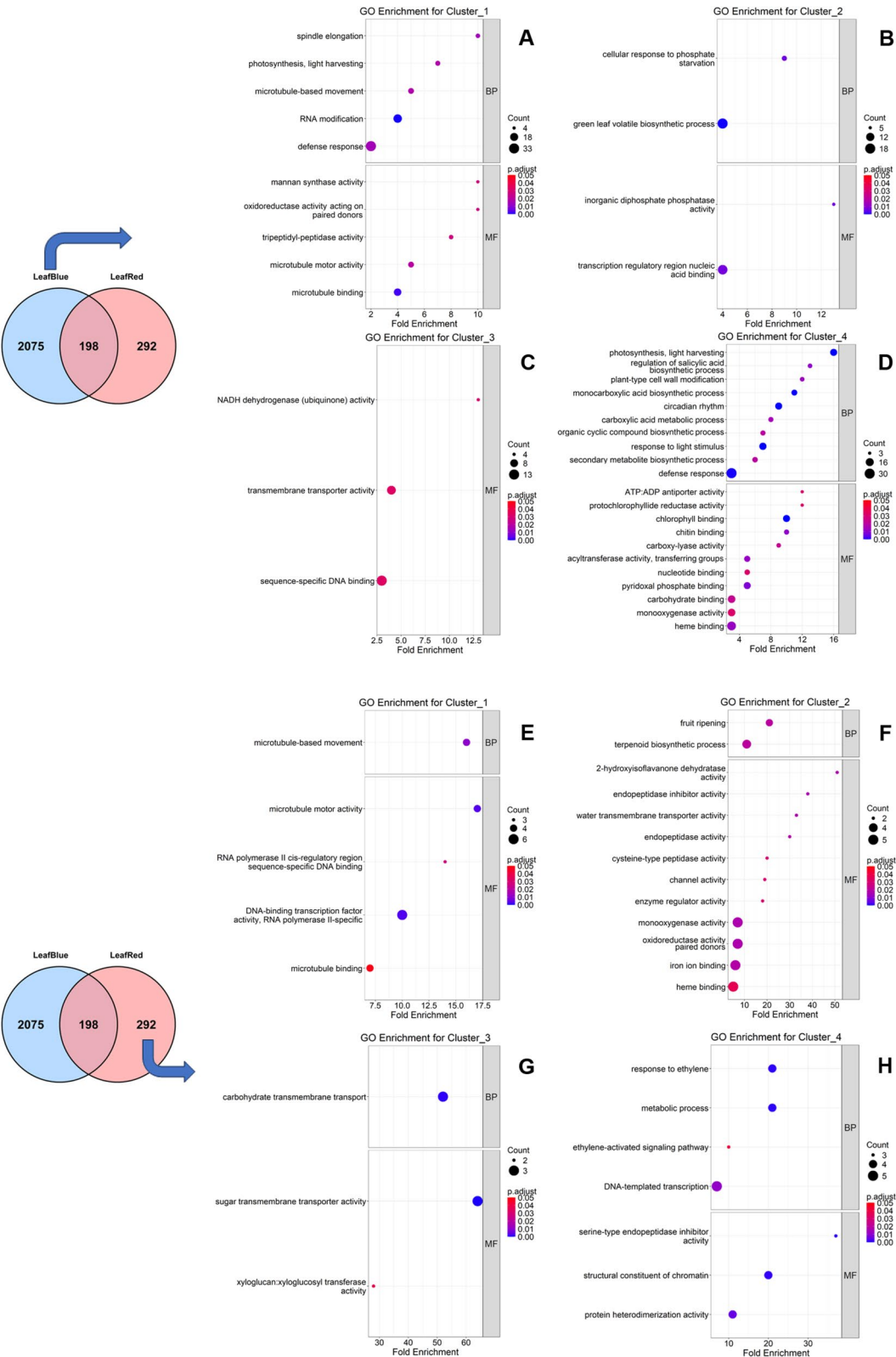


Fig. 7 Gene Ontology (GO) enrichment analysis of clusters derived from DEGs unique to Flight.Leaf.Blue vs. Ground.Leaf.Blue (A–D) and Flight.Leaf.Red vs. Ground.Leaf.Red (E–H) comparisons. The Venn diagrams on left illustrates DEGs unique to each comparison. The dot plots depict significantly enriched GO terms ($p_{\text{adjust}} < 0.05$) in biological processes (BP) and molecular function (MF) categories for each cluster

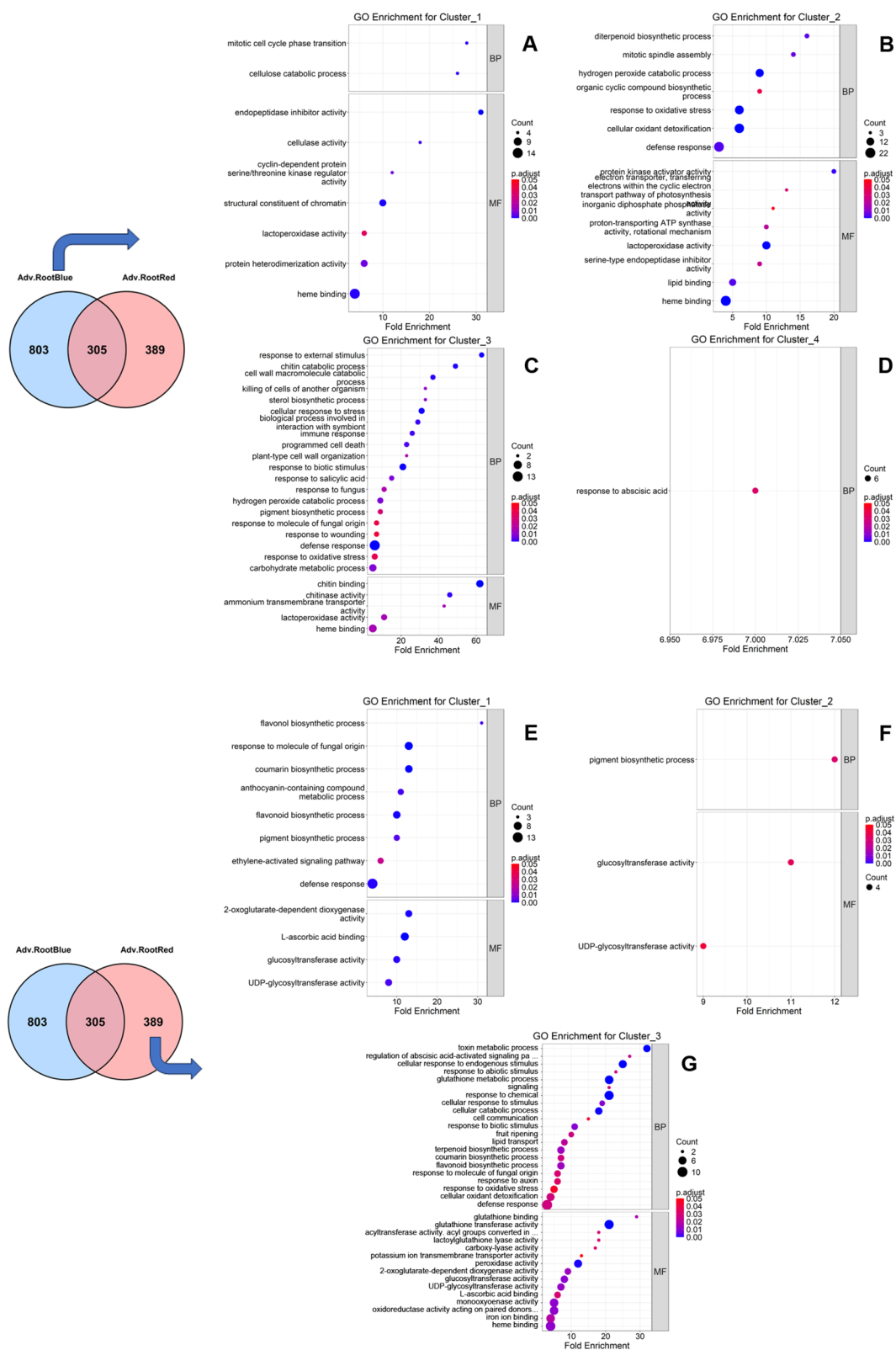


Fig. 8 Gene Ontology (GO) enrichment analysis of clusters derived from DEGs unique to Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue (A–D) and Flight.Adv.Root.Red vs. Ground.Adv.Root.Red (E–G) comparisons. The Venn diagrams on left illustrates DEGs unique to each comparison. The dot plots depict significantly enriched GO terms ($p_{\text{adjust}} < 0.05$) in biological processes (BP) and molecular function (MF) categories for each cluster

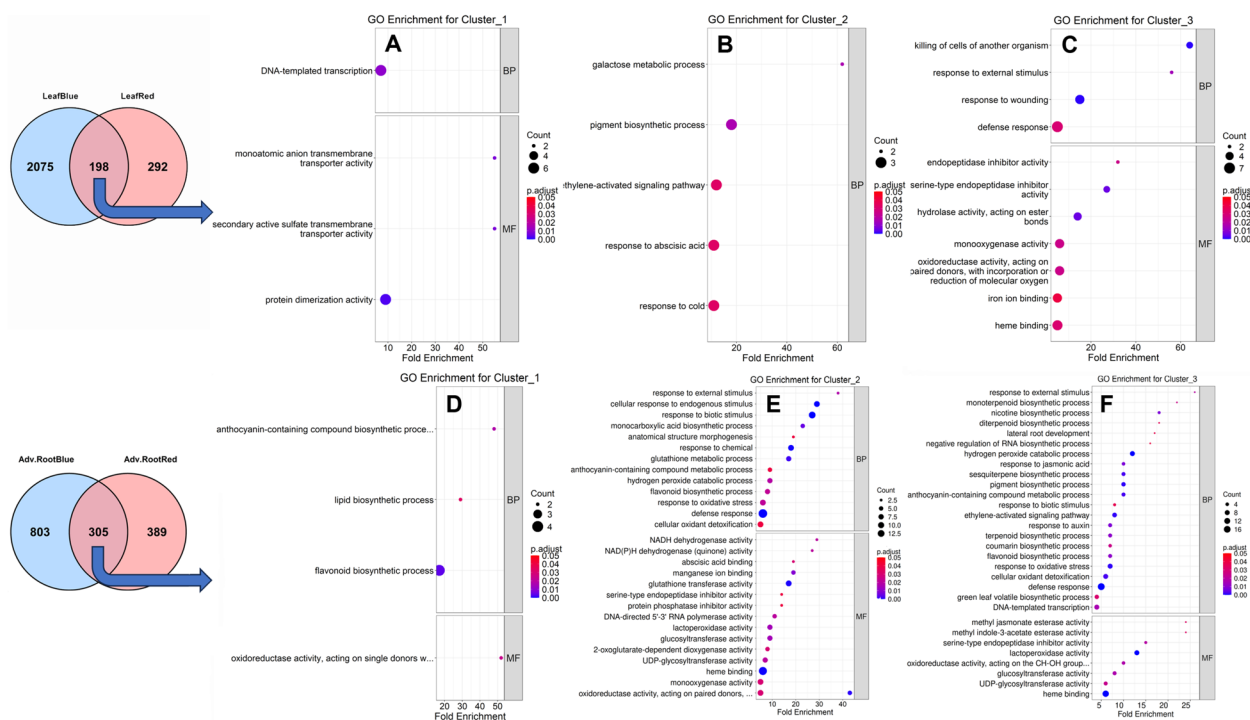


Fig. 9 Gene Ontology (GO) enrichment analysis of clusters derived from DEGs common to Flight.Leaf.Blue vs. Ground.Leaf.Blue and Flight.Leaf.Red vs. Ground.Leaf.Red (A-C), while panels (D-F) show DEGs common to Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue and Flight.Adv.Root.Red vs. Ground.Adv.Root.Red comparisons. Left-side Venn diagrams illustrate the overlap of DEGs from each respective condition. Dot plots on the right depict significantly enriched GO terms ($p_{\text{adjust}} < 0.05$) in the Biological Process (BP) and Molecular Function (MF) categories for each DEG cluster

wall metabolism, xylem development, hormonal signaling, secondary metabolite synthesis, and pigmentation, with molecular functions including chitin binding and glucan metabolism (Fig. 11A-C). GndAdv.Root exhibited enrichment for redox regulation, carbohydrate metabolism, transcriptional regulation via RNA polymerase II, hormonal signaling, and developmental processes like gibberellin response and fruit ripening (Fig. 11D-F).

For DEGs common to flight and ground leaves (45 DEGs), no significant GO terms were identified. For DEGs common to flight and ground adventitious roots (105 DEGs), GO enrichment was observed in fruit ripening, ethylene response, nitrogen metabolism, cell wall biogenesis, pectin catabolism, and enzymatic regulation, emphasizing structural and metabolic adaptations (Fig. 12A-C). This analysis underscores functional adaptations to lighting conditions and environmental differences in flight and ground-grown plants.

KEGG enrichment analyses

Flight vs. ground comparison

In LeafBlue, enriched pathways included flavonoid biosynthesis, circadian rhythm – plant, and alkaloid biosynthesis (Fig. 13A). For LeafRed, MAPK signaling, and plant hormone signal transduction pathways were most enriched (Fig. 13B). Adv.RootBlue showed significant enrichment in phenylpropanoid biosynthesis (fold

enrichment 7.37; adjusted p -value 6.41×10^{-6}), highlighting roles in lignin biosynthesis, secondary metabolism, and stress responses (Fig. 13C). No significant pathways were identified for Adv.RootRed. For DEGs common to LeafBlue and LeafRed, galactose metabolism (fold enrichment ~16.42; adjusted p -value 0.0905) was notable, suggesting conserved metabolic regulation in carbohydrate processing (Fig. 13D). For DEGs common to Adv.RootBlue and Adv.RootRed, significant pathways included flavonoid biosynthesis (fold enrichment ~21), stilbenoid and gingerol biosynthesis, alkaloid metabolism, tyrosine metabolism, and phenylpropanoid biosynthesis, underscoring shared metabolic adaptations and bioactive compound synthesis (Fig. 13E).

Blue-rich vs. red-rich comparison

In FltLeaf, enriched pathways included protein processing in the endoplasmic reticulum (fold enrichment ~5.5) and nitrogen metabolism (~5), reflecting protein folding and nitrogen utilization roles (Fig. 14A). For GndLeaf, circadian rhythm – plant (fold enrichment ~8.7; adjusted p -value 0.011) was the sole enriched pathway, emphasizing circadian-regulated physiological processes (Fig. 14B). In FltAdv.Root, pathways such as flavonoid biosynthesis (fold enrichment ~12), phenylpropanoid biosynthesis (~7.5), and stilbenoid biosynthesis (~8) were enriched, reflecting stress-responsive secondary

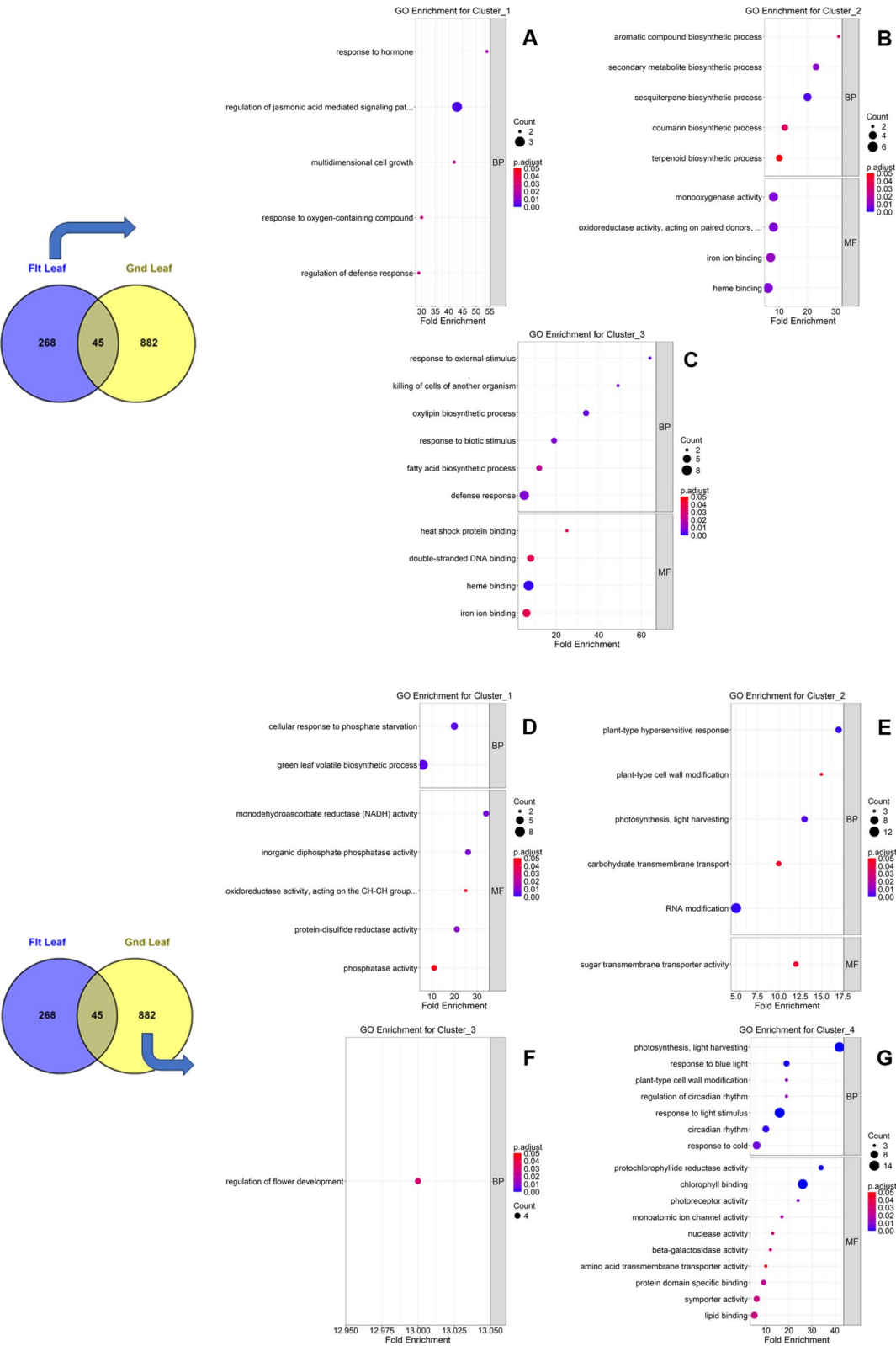


Fig. 10 Gene Ontology (GO) enrichment analysis of clusters derived from DEGs unique to Flight.Leaf.Blue vs. Flight.Leaf.Red (A-C) and Ground.Leaf.Blue vs. Ground.Leaf.Red (D-G) comparisons. The Venn diagrams on left illustrates DEGs unique to each comparison. The dot plots depict significantly enriched GO terms ($p.adjust < 0.05$) in biological processes (BP) and molecular function (MF) categories for each cluster

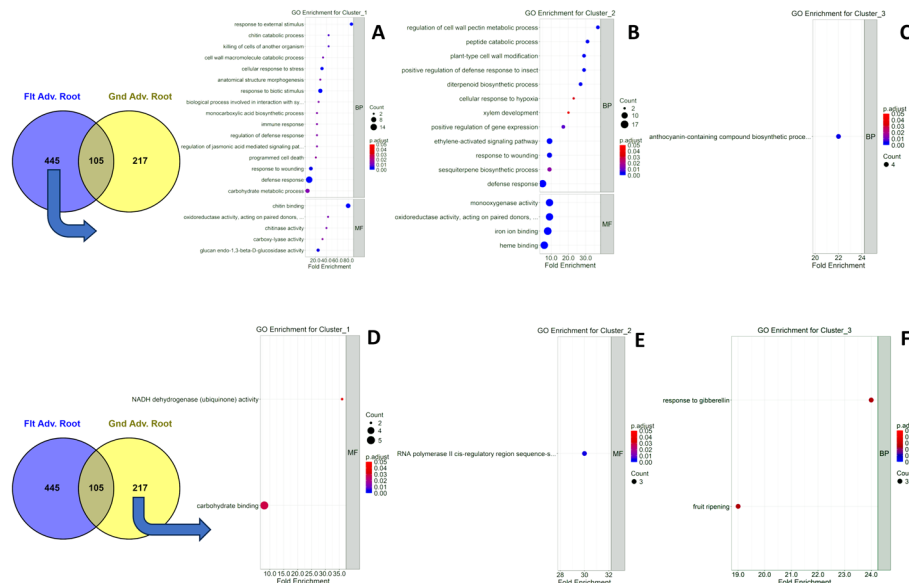


Fig. 11 Gene Ontology (GO) enrichment analysis of clusters derived from DEGs unique to Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red (A-C) and Ground.Adv.Root.Blue vs. Ground.Adv.Root.Red (D-F) comparisons. The Venn diagrams on left illustrates DEGs unique to each comparison. The dot plots depict significantly enriched GO terms (p.adjust < 0.05) in biological processes (BP) and molecular function (MF) categories for each cluster

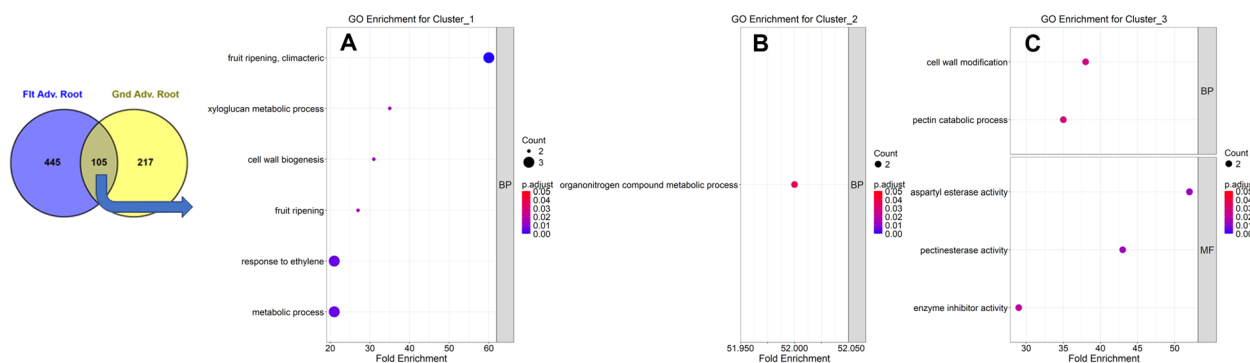


Fig. 12 Gene Ontology (GO) enrichment analysis of clusters derived from DEGs common to Flight.Leaf.Blue vs. Flight.Leaf.Red and Ground.Leaf.Blue vs. Ground.Leaf.Red (A-C) comparisons. The Venn diagrams on left illustrates DEGs unique to each comparison. The dot plots depict significantly enriched GO terms (p.adjust < 0.05) in biological processes (BP) and molecular function (MF) categories for each cluster

metabolism and lignin synthesis. Additional pathways included phenylalanine metabolism and plant hormone signal transduction (~5), highlighting hormonal regulation and metabolic precursors (Fig. 14C). No significant pathways were identified for GndAdv.Root. For DEGs common to FltLeaf and GndLeaf, enriched pathways included stilbenoid biosynthesis (fold enrichment ~35), flavonoid biosynthesis (~30), phenylalanine metabolism (~20), and plant hormone signal transduction (~10), reflecting shared adaptive mechanisms through secondary metabolism and hormonal signaling (Fig. 14D). No significant enrichment was observed for DEGs common to FltAdv.Root and GndAdv.Root. This KEGG analysis highlights pathway-specific metabolic and regulatory adaptations under flight vs. ground and blue-rich vs. red-rich conditions.

Network modules highlight tissue identity, spaceflight adaptation, and light-specific expression patterns

We used co-expression analysis to define modules of genes with similar expression patterns. We identified modules showing tissue-specific expression independent of light treatment and location. These were among the modules with the most genes (Fig. 15; module A and F). Additionally, we defined a module with genes expressed in the adventitious root, flight samples (Fig. 15; module D), but not in the respective ground controls. The GO terms enriched in this module involved antioxidant metabolism, cell wall structure and organization, and aerobic respiration, hallmarks of adventitious root growth.

Only two modules indicated differences in gene expression due to light treatment. Module C, containing 2240 genes, showed expression unique to the leaf ground controls treated with blue-rich light. Conversely, genes in

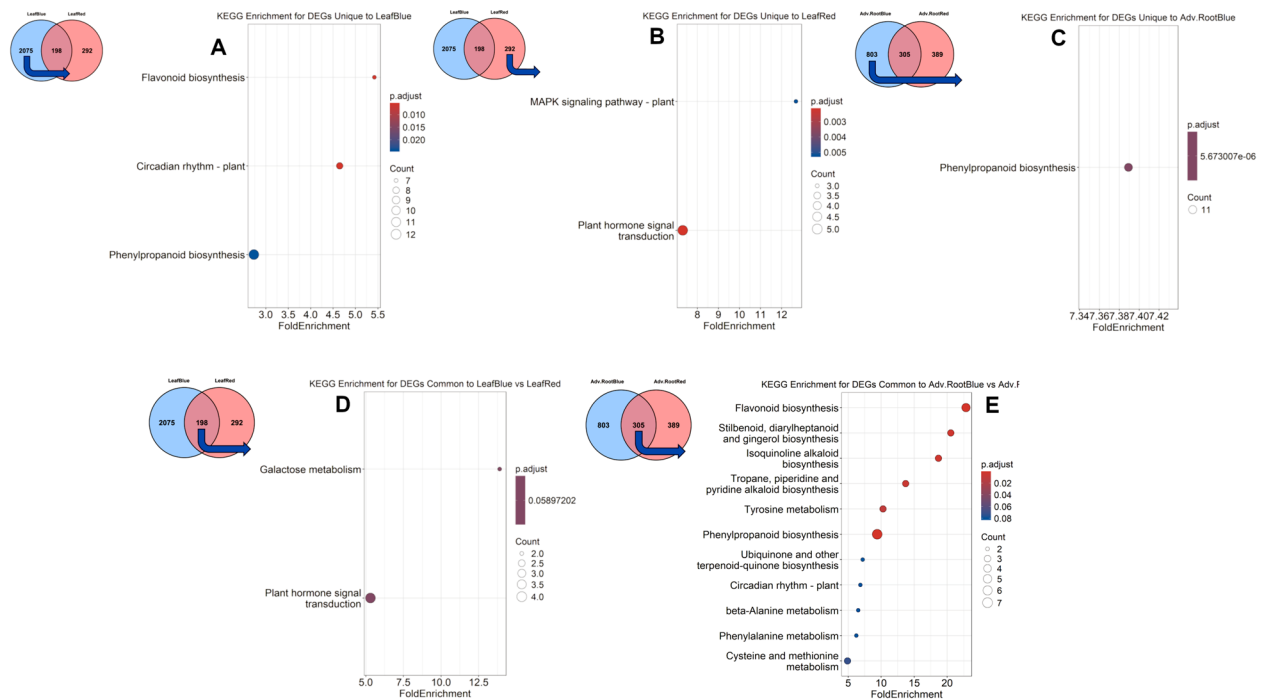


Fig. 13 KEGG pathway enrichment analysis of DEGs unique to leaves from spaceflight-grown plants under blue-rich light (Flight.Leaf.Blue vs. Ground.Leaf.Blue) (**A**), leaves from spaceflight-grown plants under red-rich light (Flight.Leaf.Red vs. Ground.Leaf.Red) (**B**), adventitious roots from spaceflight-grown plants under blue-rich light (Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue) (**C**) and common to leaves from spaceflight-grown plants under blue-rich light (Flight.Leaf.Blue vs. Ground.Leaf.Blue) vs. leaves from spaceflight-grown plants under red-rich light (Flight.Leaf.Red vs. Ground.Leaf.Red) (**D**), adventitious roots from spaceflight-grown plants under blue-rich light (Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue) vs. adventitious roots from spaceflight-grown plants under red-rich light (Flight.Adv.Root.Red vs. Ground.Adv.Root.Red) (**E**) comparisons. The Venn diagrams on left illustrates DEGs unique or common to each comparison. Dot plots illustrate significantly enriched KEGG pathways among the unique/common DEGs, with fold enrichment shown on the x-axis. Dot sizes indicate the number of genes contributing to each pathway, while colors represent the adjusted p -values ($p_{\text{adjust}} < 0.1$)

Module D are expressed in all flight and ground leaf samples except the blue-rich light treated ground controls. This suggests that stress response experience during spaceflight may hinder blue-light-specific gene expression in leaves (Fig. 15; module C and D). Although the “photosynthesis, light reactions” GO term was enriched in both modules, Module C had very few enriched GO terms and no others related to photosynthesis while many of the enriched GO terms in Module D were related to photosynthesis. Interestingly, the second largest module, Module B, containing 4166 genes expressed in the adventitious roots of the ground controls, but had very inconsistent gene expression in both the leaves and adventitious roots of the flight samples (Fig. 15 module B). Many of the enriched GO terms in this module were related to cellular transport and exocytosis.

Discussion

Biological context and significance

The findings of this study underscore how spaceflight conditions and spectral light quality interact to reshape plant transcriptional programs in a species- and tissue-specific manner. While prior spaceflight transcriptomic studies, particularly those in *Arabidopsis*, have

consistently identified activation of oxidative stress responses, hormonal signaling networks (notably auxin, ABA, and JA), and circadian disruption, our results extend this understanding into a fruiting crop with complex developmental physiology. The tomato transcriptome revealed not only conserved stress adaptations but also unique shifts in metabolic and structural pathways, particularly in adventitious roots. These roots exhibited strong transcriptional signatures for secondary metabolite biosynthesis, lipid remodeling, and nutrient transporter activation traits less prominent or absent in *Arabidopsis* datasets [52, 53]. Notably, this aligns with findings that emphasize the impact of microgravity on the reproductive success and metabolic pathways of tomato plants, highlighting the complexity of developmental physiology in microgravity contexts [54].

Moreover, the light-dependent variability in tomato gene expression, particularly under blue-rich conditions in space, emphasizes that fruiting crops may rely on broader regulatory architectures to manage environmental stress. Whereas *Arabidopsis* leaves often downregulate photosynthesis-related genes and plastid-associated transcriptional machinery in microgravity, tomato leaf tissues maintained or even enhanced expression of

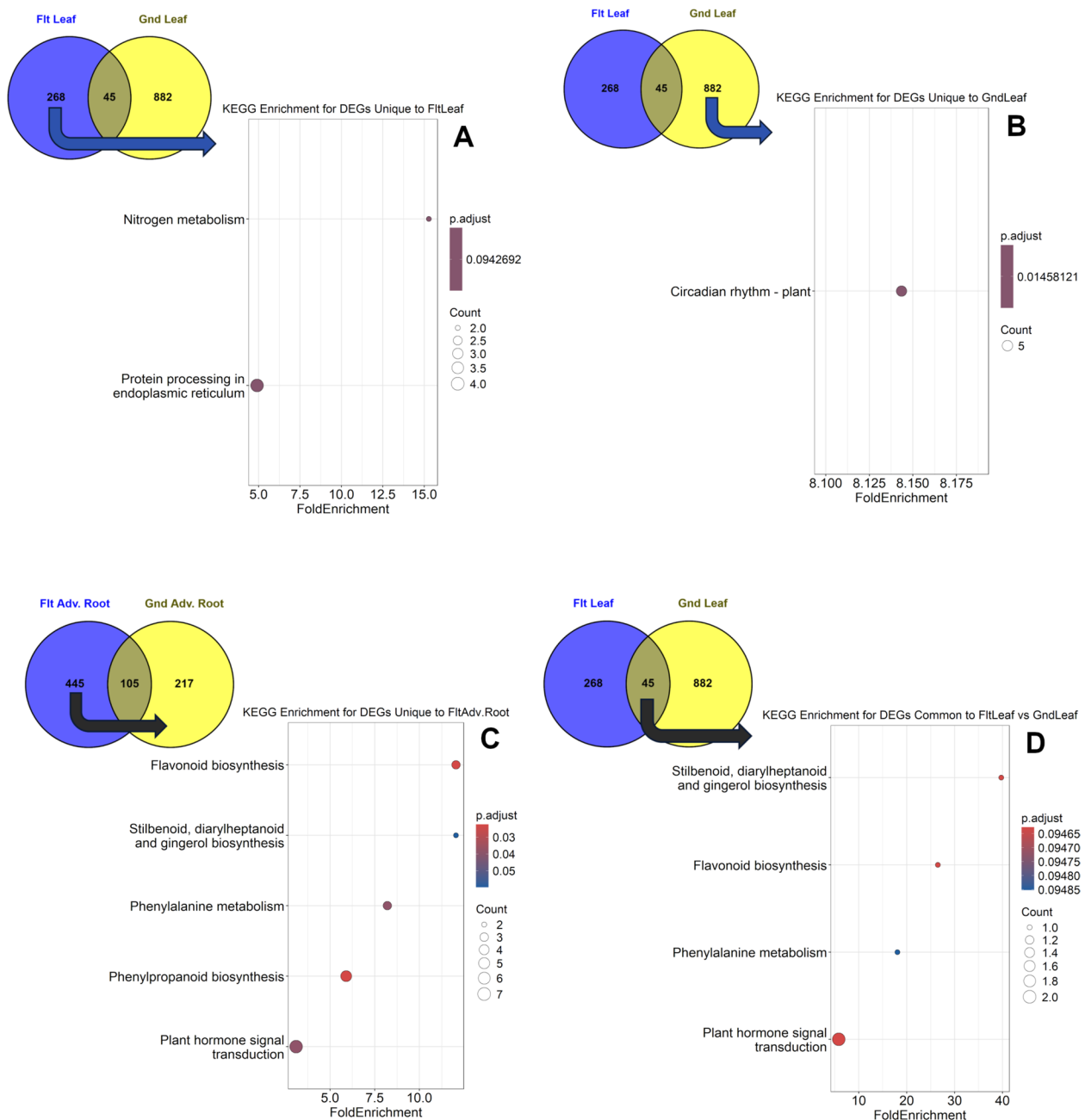


Fig. 14 KEGG pathway enrichment analysis of clusters derived from DEGs unique to Flight.Leaf.Blue vs. Flight.Leaf.Red comparison (**A**), Ground.Leaf.Blue vs. Ground.Leaf.Red comparison (**B**), Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red comparison (**C**) and common to Flight.Leaf.Blue vs. Flight.Leaf.Red comparison vs. Ground.Leaf.Blue vs. Ground.Leaf.Red comparison (**D**) comparisons. The Venn diagrams on left illustrates DEGs unique or common to each comparison. Dot plots illustrate significantly enriched KEGG pathways among the unique/common DEGs, with fold enrichment shown on the x-axis. Dot sizes indicate the number of genes contributing to each pathway, while colors represent the adjusted p -values ($p.adjust < 0.1$)

chloroplast function genes under red-rich light. This suggests not only greater transcriptomic resilience in tomato but also potentially divergent photoreceptor or retrograde signaling dynamics compared to model systems. This observation is consistent with previous research indicating that photoreception under different light spectra can significantly alter metabolic pathways in plants [50, 55]. Taken together, these results demonstrate

that spaceflight elicits both conserved and crop-specific molecular adaptations, supporting the necessity of expanding beyond Arabidopsis in space biology studies.

Spectral light responses under normal gravity

Under normal Earth gravity, red and blue light modulate distinct physiological and molecular processes. Blue light is known to enhance chloroplast biogenesis, stomatal

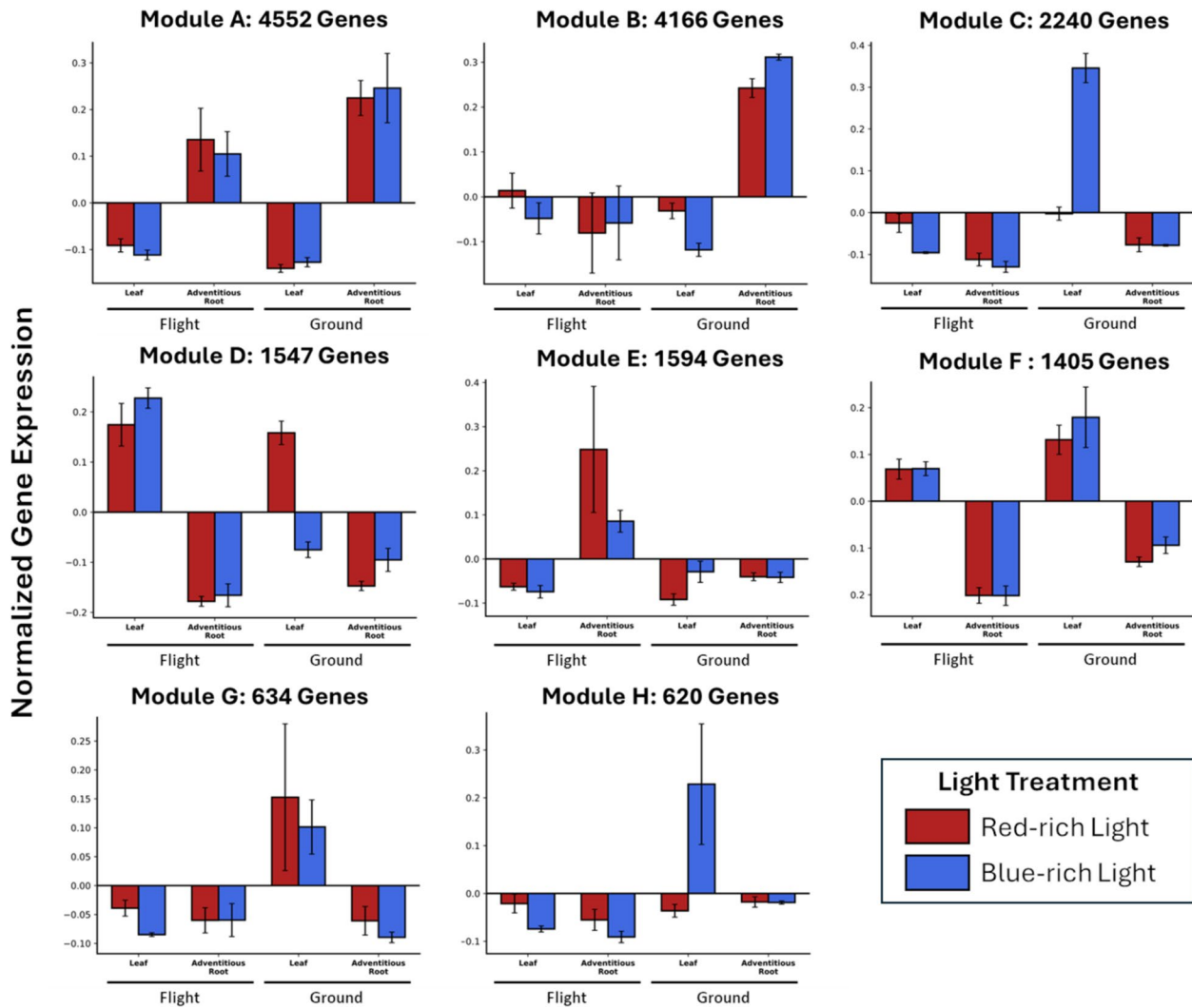


Fig. 15 Eigengene expression profiles for WGCNA modules (A–H). Each panel displays the normalized eigengene expression values for a co-expression module across different sample groups, separated by tissue type (leaf, adventitious root), light treatment (red-rich, blue-rich), and growth condition (flight, ground). Bars represent the average eigengene expression for each group, with error bars indicating standard error. The number of genes in each module is noted in the panel title. Red and blue bars correspond to red-rich and blue-rich light treatments, respectively

opening, and the activation of antioxidant defense systems through cryptochrome and phototropin signaling pathways [55]. Red light, perceived by phytochromes, drives photosynthetic competence and maintains metabolic homeostasis, influencing seedling photomorphogenesis, shade avoidance, and circadian entrainment [56, 57]. Our ground control data are consistent with these established paradigms: blue-rich lighting enriched gene expression related to oxidative stress response, hormone signaling (notably ABA), and detoxification pathways, while red-rich lighting fostered stability in genes associated with primary metabolism, structural maintenance, and chloroplast function. In our own Earth-grown ground control samples, RNA-seq comparisons between red and blue light treatments further reinforced these findings.

It is worth noting why multiple clustering approaches were applied in this study. Global clustering across all samples primarily separated tissues or light conditions, but this broad structure masked subtler spaceflight $\times$ light interactions. To capture these condition-specific effects, we clustered DEGs from individual pairwise contrasts (e.g., LeafBlue: Flight vs. Ground). This approach uncovered distinct transcriptional programs such as flight-induced stress signaling under blue light—that global analyses did not resolve. In combination with WGCNA, which examined higher-order co-expression across all samples, this dual strategy provided both fine-grained resolution and broader network-level insights.

Implications of blue-rich light induced variability

Our analyses consistently showed that blue-rich light amplified variability in transcriptomic responses to spaceflight, whereas red-rich light constrained expression changes. This pattern was supported not only by DEG counts but also by variance metrics, overlap analyses, and fold-change magnitudes (see Results; Supplementary Figs. 13–17). Together, these findings indicate that red light actively stabilizes transcriptional programs, while blue light promotes dynamic, condition-sensitive responses a distinction likely tied to photoreceptor signaling and its modulation in microgravity.

Photoreceptor pathway perturbations in space

Spaceflight has been shown to alter photoreceptor-mediated physiological responses. For example, red-light and blue-light phototropism exhibit altered patterns in *Arabidopsis* seedling roots and hypocotyls under microgravity aboard the ISS [58, 59]. Although direct evidence for changes in photoreceptor localization such as phytochrome B nuclear body dynamics or cryptochrome nuclear import is lacking, these phototropic anomalies suggest that light signaling pathways may be functionally impaired in microgravity. In our dataset, Weighted Gene Co-expression Network Analysis (WGCNA) revealed that gene modules enriched in blue-light-responsive ground samples were absent or highly disrupted in flight, suggesting that microgravity may interfere with the transcriptional coordination normally mediated by blue light pathways. Although CRY and PHOT photoreceptor transcripts did not show strong differential expression, the altered regulation of downstream targets such as HY5 homologs and flavonoid biosynthetic genes suggests that photoreceptor signaling may be modulated at the level of downstream transcriptional networks during spaceflight. No current spaceflight study has directly demonstrated transcriptional or post-translational regulation of HY5 or COP1, yet their central roles in light response pathways are well established in *Arabidopsis*. Under Earth conditions, COP1 targets HY5 for proteasomal degradation in darkness, while blue light inhibits COP1 activity, thereby stabilizing HY5 and promoting photomorphogenesis [58, 59]. Therefore, while our findings suggest a potential disruption in this regulatory axis under microgravity, such interpretations remain speculative until validated through targeted functional studies. These observed pathway disturbances, coupled with transcriptomic variability under blue-rich light, reinforce the hypothesis that altered gravity perturbs photoreceptor signaling and downstream gene coordination. This underscores the need for mechanistic investigations into photoreceptor localization, signaling fidelity, and nuclear trafficking in the spaceflight environment.

Insights into adventitious root formation in space

A key phenotypic observation from this study was the pronounced adventitious root formation in space-grown tomatoes, which was substantially greater than in ground controls. Our transcriptomic data offer mechanistic insights into this phenomenon. Adventitious root formation under terrestrial conditions is commonly induced by stressors such as flooding, hypoxia, and nutrient deprivation, all of which converge on hormonal pathways especially those involving auxin, ethylene, and cytokinin [60, 61]. In our dataset, adventitious root tissues from space-grown plants exhibited strong upregulation of genes involved in auxin transport (e.g., PINs, AUX/LAX family members), ethylene biosynthesis (ACS, ACO), and response pathways, as well as reactive oxygen species (ROS) detoxification and cell wall modification enzymes such as expansins and peroxidases. Notably, blue-rich light in flight conditions amplified these responses, with enhanced expression of stress-related transcription factors (e.g., WRKYs, NACs), phenylpropanoid biosynthetic genes, and glutathione-S-transferases, pointing toward an integrated response to environmental cues. KEGG and GO enrichment further highlighted flavonoid biosynthesis, ABA signaling, and oxidative stress response pathways as central players in adventitious root formation under microgravity. These results suggest that spaceflight imposes physiological stressors likely related to root zone oxygenation, fluid distribution, and altered gravitropic signaling that converge on conserved stress-induced developmental programs, triggering adventitious roots development as a compensatory adaptation. Previous studies confirm the multifactorial traits involved in adventitious rooting influenced by hormonal interplay and oxidative stress in microgravity [60, 62]. This study thus reinforces the concept that adventitious root formation in microgravity is a complex, multifactorial phenotypic response involving hormonal cross-talk, redox regulation, and structural remodeling. These findings provide not only an explanation for the excess adventitious roots formation observed in flight-grown tomatoes but also a foundation for exploring genetic or environmental strategies to manage root architecture in extraterrestrial cultivation systems.

Integration with arabidopsis ISS datasets

Recent studies highlight the significance of analyzing transcriptional responses in plants, particularly in the context of microgravity environments like those encountered in space. Comparative assessments across different plant species, notably *Arabidopsis* and tomato, reveal critical insights into their adaptive strategies under these conditions. Research indicates that gene expression modifications are essential for plants' survival in space, with divergent responses noted between species. For instance,

Arabidopsis demonstrates considerable transcriptional shifts in response to spaceflight stresses, often affecting broad stress-associated pathways across tissues. By contrast, tomato shows transcriptional changes that, while equally significant, are more tissue-specific and pronounced in specialized organs such as adventitious roots, highlighting its species-specific adaptive strategies [63–65]. In the context of data derived from the International Space Station (ISS), two key datasets the APEX03-2 [66, 67] and VEGGIE-APEX [68] serve as invaluable resources for understanding the core pathways of stress adaptation in plants. These datasets indicate that oxidative stress responses, particularly those involving glutathione metabolism and peroxidase activity, alongside hormone signaling pathways (e.g., abscisic acid and jasmonic acid), are fundamentally altered in microgravity [52, 69, 70]. Studies indicate that *Arabidopsis* stress resilience is enhanced through modifications in hormonal pathways and gene expressions associated with drought and oxidative stress [71, 72]. However, tomato exhibits more extensive transcriptional reprogramming, particularly under specialized conditions, showcasing gene activities tied to cell wall biosynthesis and nutrient transport, which are less prevalent in *Arabidopsis* datasets [65, 73]. The anatomical differences between rosette herbaceous plants like *Arabidopsis* and fruiting crops like tomato lead to their respective behavioral adaptations under spaceflight challenges [74]. Moreover, investigations into ecotypic variations within *Arabidopsis* reveal a broader array of genetic backgrounds contributing to differing stress responses a finding critical for breeding more resilient crop varieties capable of thriving in extreme conditions [75]. Comparatively, tomato's adaptive responses, which include modifications in structures such as adventitious roots exposed to particular light spectra, emphasize its versatility to allocate biomass effectively to reproductive structures during stress [58]. This ability to dynamically adjust transcription in response to environmental stimuli is pivotal for the success of plant cultivation during space missions [76, 77]. The intricate interplay of transcriptional changes in response to microgravity not only underscores the species-specific resilience mechanisms but also emphasizes the opportunities for genetic enhancement to improve plant adaptations. Thus, further research integrating the findings from ISS-grown *Arabidopsis* and tomato can shed light on optimizing agricultural practices for sustainable food production in space habitats [50].

Implications for spaceflight agriculture and crop resilience

The combined effects of light spectral quality and microgravity emerge as critical determinants of plant adaptive responses in spaceflight environments. Our findings offer two practical implications. First, red-rich lighting

was associated with fewer and less variable transcriptional changes between flight and ground conditions, suggesting it may stabilize gene expression and promote more predictable physiological outcomes during space cultivation. Incorporating red-rich light into bioregenerative life-support systems may therefore help reduce stress-induced variability and enhance crop management in long-duration missions. Second, the pronounced adventitious root formation observed under microgravity likely driven by altered auxin signaling and uneven water distribution underscores the need for precise root-zone moisture control. Addressing this challenge will be essential for maintaining healthy root development and minimizing unintended stress responses. Beyond spaceflight, insights into oxidative stress tolerance, hormonal plasticity, and nutrient transport gleaned from this study may inform breeding strategies for climate-resilient crops and improve the design of vertical farming systems on Earth.

Conclusion

This study presents a detailed transcriptional landscape of *Solanum lycopersicum* cv. Red Robin grown under spaceflight conditions and varying light spectra. It explores the interplay between photoreception, stress responses, and developmental processes, with particular focus on adventitious root formation and the effects of blue-rich light signaling in microgravity. By bridging the knowledge gap between *Arabidopsis*-centric space biology and fruit crop systems, this research enhances our understanding of plant stress adaptation in space and offers insights into the potential for sustainable agriculture in extreme environments, paving the way for future food production beyond Earth.

Abbreviations

°C	Degrees Celsius
ATP	Adenosine triphosphate
BP	Biological process
CO ₂	Carbon dioxide
CRF	Controlled release fertilizer
CRISPR-Cas9	Clustered regularly interspaced short palindromic repeats
CRS	Commercial resupply service
DEG	Differentially expressed genes
DNA	Deoxyribonucleic acid
FDR	False discovery rates
GO	Gene ontology
HOB0	Honest observer by onset
HRP	Human research program
ILSR	International life sciences research announcement
IMA	Inhibitory mold agar
ISS	International space station
KEGG	Kyoto encyclopedia of genes and genomes
KSC	Kennedy space center
LED	Light emitting diode
MAPK	Mitogen-activated protein kinase
MELFI	Minus eighty laboratory freezer for ISS
MF	Molecular function
mRNA	Messenger ribonucleic acid
NADH	Nicotinamide adenine dinucleotide
NADPH	Nicotinamide adenine dinucleotide phosphate
NCBI	National center for biotechnology information

NRA	NASA research announcement
OSDR	Open science data repository
PAR	Photosynthetic active radiation
PCA	Principal component analysis
PPC	Profile porous ceramics
ppm	Parts per million
RH	Relative humidity
RNA	Ribonucleotide acid
ROS	Reactive oxygen species
TSA	Tryptic soy agar
WGCNA	Weighted gene co-expression network analysis

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-07621-4>.

Supplementary Material 1: Supplemental Fig. 1: Heatmaps showing expression patterns for clusters (A-D) derived from DEGs unique to the LeafBlue (Flight.Leaf.Blue vs. Ground.Leaf.Blue) comparison (see Fig. 3A & 5A). Each panel (A-D) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 2: Supplemental Fig. 2: Heatmaps showing expression patterns for clusters (A-D) derived from DEGs unique to the LeafRed (Flight.Leaf.Red vs. Ground.Leaf.Red) comparison (see Fig. 3A & 5B). Each panel (A-D) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 3: Supplemental Fig. 3: Heatmaps showing expression patterns for clusters (A-D) derived from DEGs unique to the Adv.RootBlue (Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue) comparison (see Fig. 3C & 5C). Each panel (A-D) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 4: Supplemental Fig. 4: Heatmaps showing expression patterns for clusters (A-C) derived from DEGs unique to the Adv.RootRed (Flight.Adv.Root.Red vs. Ground.Adv.Root.Red) comparison (see Fig. 3C & 5D). Each panel (A-C) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 5: Supplemental Fig. 5: Heatmaps showing expression patterns for clusters (A-C) derived from DEGs common to the LeafBlue_LeafRed (common to Flight.Leaf.Blue vs. Ground.Leaf.Blue and Flight.Leaf.Red vs. Ground.Leaf.Red) comparison (see Fig. 3A & 5E). Each panel (A-C) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 6: Supplemental Fig. 6: Heatmaps showing expression patterns for clusters (A-C) derived from DEGs common to the Adv.RootBlue_Adv.RootRed (common to Flight.Adv.Root.Blue vs. Ground.Adv.Root.Blue and Flight.Adv.Root.Red vs. Ground.Adv.Root.Red) comparison (see Fig. 3C & 5F). Each panel (A-C) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-

average expression.

Supplementary Material 7: Supplemental Fig. 7: Heatmaps showing expression patterns for clusters (A-D) derived from DEGs unique to the FltLeaf (Flight.Leaf.Blue vs. Flight.Leaf.Red) comparison (see Fig. 4A & 6A). Each panel (A-D) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 8: Supplemental Fig. 8: Heatmaps showing expression patterns for clusters (A-D) derived from DEGs unique to the GndLeaf (Ground.Leaf.Blue vs. Ground.Leaf.Red) comparison (see Fig. 4A & 6B). Each panel (A-D) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 9: Supplemental Fig. 9: Heatmaps showing expression patterns for clusters (A-C) derived from DEGs unique to the FltAdv.Root (Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red) comparison (see Fig. 4C & 6C). Each panel (A-C) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 10: Supplemental Fig. 10: Heatmaps showing expression patterns for clusters (A-C) derived from DEGs unique to the GndAdv.Root (Ground.Adv.Root.Blue vs. Ground.Adv.Root.Red) comparison (see Fig. 4C & 6D). Each panel (A-C) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 11: Supplemental Fig. 11: Heatmaps showing expression patterns for clusters (A-C) derived from DEGs common to the FltLeaf_GndLeaf (Flight.Leaf.Blue vs. Flight.Leaf.Red and Ground.Leaf.Blue vs. Ground.Leaf.Red) comparison (see Fig. 4C & 6E). Each panel (A-C) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 12: Supplemental Fig. 12: Heatmaps showing expression patterns for clusters (A-C) derived from DEGs common to the FltAdv.Root_GndAdv.Root (Flight.Adv.Root.Blue vs. Flight.Adv.Root.Red and Ground.Adv.Root.Blue vs. Ground.Adv.Root.Red) comparison (see Fig. 4C & 6F). Each panel (A-C) corresponds to a distinct k-means cluster, with rows representing genes and columns representing samples. Expression values are based on log₂-transformed CPM (log₂CPM) and scaled by row (z-score) to highlight relative differences across samples. Red indicates higher-than-average expression, blue indicates lower-than-average expression.

Supplementary Material 13: Supplemental Fig. 13: (A) Principal Component Analysis (PCA) of logCPM expression data across flight and ground samples under red- and blue-rich light conditions. (B) Violin and boxplots showing the distribution of gene-wise coefficient of variation (CV) under red and blue light treatments.

Supplementary Material 14: Supplemental Fig. 14: Boxplots showing logCPM expression of genes with significant Light × Condition interaction effects, across flight and ground samples under red- and blue-rich lighting.

Supplementary Material 15: Supplemental Fig. 15: Venn diagram showing the overlap of differentially expressed genes (DEGs) across four comparisons: Red-Flight vs. Red-Ground, Blue-Flight vs. Blue-Ground, Red-Ground vs. Blue-Ground, and Red-Flight vs. Blue-Flight.

Supplementary Material 16: Supplemental Fig. 16: UpSet plot illustrating

the intersection of DEG sets across four comparisons: Red-Flight vs. Red-Ground, Blue-Flight vs. Blue-Ground, Red-Ground vs. Blue-Ground, and Red-Flight vs. Blue-Flight.

Supplementary Material 17: Supplemental Fig. 17: Barplots showing (A) mean absolute log2 fold change ($|\log_2FC|$) and (B) cumulative absolute log2 fold change across DEG sets for each comparison.

Supplementary Material 18: Supplemental Fig. 18: Principal Component Analysis (PCA) of all tomato RNA-seq samples across tissues, light treatments, and flight conditions. (A) PCA of all samples, with tissue type indicated by color (black = adventitious root, orange = leaf), condition by shape (circle = flight, square = ground), and light treatment by fill colors (blue = blue-rich, red = red-rich). Tissue identity accounted for the majority of variance (PC1), while condition and light quality explained additional structure along PC2. (B) The same PCA results faceted by light treatment (blue-rich vs. red-rich) to highlight within-light differences in transcriptional responses between flight and ground.

Supplementary Material 19: Supplemental Table 1: Sequencing and mapping statistics for tomato samples grown under red-rich or blue-rich light in spaceflight and ground control conditions. Values include raw read counts (R1+R2), percentage of reads retained post-quality trimming, and STAR alignment outcomes: uniquely mapped reads, multi-mapped reads, reads unmapped due to short length, and reads unmapped for other reasons.

Supplementary Material 20: Supplemental File 1: Silhouette Clusters for All Comparisons. This file contains the results of k-means clustering analyses performed on differentially expressed genes (DEGs) across all pairwise comparisons (see Table 3 for abbreviations). Each worksheet corresponds to a specific comparison (e.g., LeafBlue, LeafRed, Adv.RootBlue, etc.), with genes assigned to clusters based on silhouette-validated patterns of expression. Columns include gene identifiers (ITAG4.1), assigned cluster numbers, average Silhouette widths, and associated metadata.

Supplementary Material 21: Supplemental File 2: Complete GO enrichment results for all clusters described in the manuscript. Each zip compressed file corresponds to one dataset (e.g., LeafBlue, Adv.RootRed, etc.), and each file inside uncompressed folder provides results for individual clusters. Columns include GO term ID, description, ontology category, gene ratio, fold enrichment, adjusted p-value, and contributing gene IDs.

Supplementary Material 22: Supplemental File 3: R Script for Cluster Annotation and Functional Group Integration. R script used to generate functional group summaries for Silhouette-clustered DEGs. The script integrates expression data, cluster assignments, DEG statistics, and pre-computed GO enrichment results. Functional categories were derived from ITAG4.1 annotations and condensed GO terms, while representative genes were selected based on expression strength and statistical significance. This script ensures transparency and reproducibility of the functional group assignments and cluster summaries presented in the main tables and figures.

Acknowledgements

The authors would like to thank the astronauts aboard ISS during Expedition 68 for their diligence in maintaining and caring for these crops during this flight experiment. In addition to the ISS astronaut team, the authors would also like to extend their gratitude to all of those in the Veggie ground support teams involved including the COMET and AEGIS teams and personnel at the Payload Operations and Integration Center for pre-flight preparations, and support of the flight experiment and ground control experiment. We would like to thank team members Dr. Robert Ploutz-Snyder and Dr. Millenia Young for their statistical advice and Dr. Grace Douglas for her support with all aspects of this research grant. We are grateful for the help of numerous undergraduate and graduate student interns at Kennedy Space Center and our partners for their help during sample processing. The authors most definitely would like to extend their appreciation to Mr. Matthew Romeyn and Ms. Jess Bunche for their efforts during the pre-flight crop growth tests and Dr. Ed Rosenthal for fertilizer support. We would also like to extend our gratitude to NASA and contractor scientists for their helpful discussions and planning including, Mr. Jeff Richards and Dr. Natasha Haveman as well as Project Scientist Dr. Ye Zhang and Project Managers Mr. Trent Smith, Ms.

Dinah Dimapilis, and Ms. Lucy Orozco. We would like to thank NASA's ISS, Space Biology, and Human Research Programs for their support during these endeavors. We are also grateful to the students who helped us develop the VEG-05 mission patch as part of the Fairchild Tropical Botanic Garden's National Challenge.

Authors' contributions

ARD, CLK, LES, RCM, RMW, CAM, GDM – Contributed to the Experimental Design; ARD, CLK, JMS, Data Analysis; ARD, CLK, JMS – Contributed to the Manuscript Development; all authors reviewed the manuscript for accuracy.

Funding

This research was co-funded by NASA's Human Research Program and Space Biology Program as part of the ILSRA 2015 NRA submission "Pick-and-eat salad-crop productivity, nutritional value, and acceptability to supplement the ISS food system".

Data availability

The datasets generated and/or analyzed during the current study are available in the NASA Open Science Data Repository (OSDR), Project # OSD-767, <https://doi.org/https://doi.org/10.26030/rtrk-h481>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹AETOS Systems Inc., Kennedy Space Center, Orlando, FL, USA

²Noetic Strategies Inc., Kennedy Space Center, Orlando, FL, USA

³Department of Plant Biology, Plant Biology Laboratories, Michigan State University, 612 Wilson Rd #166, East Lansing, MI 48824, USA

⁴Sierra Space Inc., Madison, WI, USA

⁵Department of Horticulture and Landscape Architecture, Purdue University, West Lafayette, IN 4790, USA

⁶Exploration Research & Technology (ER&T) Programs, NASA Emeritus, Kennedy Space Center, Orlando, FL 32899, USA

Received: 23 December 2024 / Accepted: 20 October 2025

Published online: 07 April 2026

References

1. Atkinson NJ, Urwin PE. The interaction of plant biotic and abiotic stresses: from genes to the field. *J Exp Bot*. 2012;63:3523–43. <https://doi.org/10.1093/jxb/ers100>.
2. Nguyen D, Rieu I, Mariani C, Van Dam NM. How plants handle multiple stresses: hormonal interactions underlying responses to abiotic stress and insect herbivory. *Plant Mol Biol*. 2016;91:727–40. <https://doi.org/10.1007/s1103-016-0481-8>.
3. Urano K, Kurihara Y, Seki M, Shinozaki K. 'Omics' analyses of regulatory networks in plant abiotic stress responses. *Curr Opin Plant Biol*. 2010;13:132–8. <https://doi.org/10.1016/j.pbi.2009.12.006>.
4. De Pascale S, Arena C, Aronne G, De Micco V, Pannico A, Paradiso R, et al. Biology and crop production in space environments: challenges and opportunities. *Life Sci Space Res*. 2021;29:30–7. <https://doi.org/10.1016/j.lssr.2021.02.005>.
5. Reed DW, Vanden Bosch CA. Engineering perspectives of growing plants in space. *J Indian Inst Sci*. 2023;103:797–805. <https://doi.org/10.1007/s41745-023-00369-6>.
6. Maffei ME, Balestrini R, Costantino P, Lanfranco L, Morgante M, Battistelli A, et al. The physiology of plants in the context of space exploration. *Commun Biol*. 2024;7:1–12. <https://doi.org/10.1038/s42003-024-06989-7>.

7. Maity T, Saxena A. Challenges and innovations in food and water availability for a sustainable Mars colonization. *Life Sci Space Res.* 2024;42:27–36. <https://doi.org/10.1016/j.lssr.2024.03.008>.
8. Wheeler RM. Plants for human life support in space: from Myers to Mars. *Gravitational and Space Biology.* 2010;23:25–36.
9. Nguyen MTP, Knowing M, Tran NN, Burgess A, Fisk I, Watt M, et al. Space farming: horticulture systems on spacecraft and outlook to planetary space exploration. *Plant Physiol Biochem.* 2023;194:708–21. <https://doi.org/10.1016/j.plaphy.2022.12.017>.
10. Khodadad CLM, Hummerick ME, Spencer LSE, Dixit AR, Richards JT, Romeyn MW, et al. Microbiological and nutritional analysis of lettuce crops grown on the international space station. *Front Plant Sci.* 2020;11:199. <https://doi.org/10.3389/fpls.2020.00199>.
11. Hummerick ME, Khodadad CLMM, Dixit AR, Spencer LE, Maldonado-Vasquez GJ, Gooden JL, et al. Spatial characterization of microbial communities on multi-species leafy greens grown simultaneously in the vegetable production systems on the international space station. *Life.* 2021;11:1060.
12. Bunchek JM, Hummerick ME, Spencer LE, Romeyn MW, Young M, Morrow RC, et al. Pick-and-eat space crop production flight testing on the International Space Station. *J Plant Interact.* 2024;19:2292220. <https://doi.org/10.1080/17429145.2023.2292220>.
13. Douglas GL, Wheeler RM, Fritsche RF. Sustaining astronauts: resource limitations, technology needs, and parallels between spaceflight food systems and those on earth. *Sustainability.* 2021;13:9424. <https://doi.org/10.3390/su13169424>.
14. Mortimer JC, Gilliam M. Spacehort: redesigning plants to support space exploration and on-earth sustainability. *Curr Opin Biotechnol.* 2022;73:246–52. <https://doi.org/10.1016/j.copbio.2021.08.018>.
15. Sugimoto M, Oono Y, Gusev O, Matsumoto T, Yazawa T, Levinskikh MA, et al. Genome-wide expression analysis of reactive oxygen species gene network in Mizuna plants grown in long-term spaceflight. *BMC Plant Biol.* 2014;14:4. <https://doi.org/10.1186/1471-2229-14-4>.
16. Massa GD, Dufour NF, Carver JA, Hummerick ME, Wheeler RM, Morrow RC, et al. VEG-01: Veggie hardware validation testing on the international space station. *Open Agric.* 2017;2:33–41. <https://doi.org/10.1515/opag-2017-0003>.
17. Sychev VN, Levinskikh MA, Gostimsky SA, Bingham GE, Podolsky IG. Space-flight effects on consecutive generations of peas grown onboard the Russian segment of the International Space Station. *Acta Astronaut.* 2007;60:426–32. <https://doi.org/10.1016/j.actaastro.2006.09.009>.
18. Zabel P, Bamsey M, Schubert D, Tajmar M. Review and analysis of over 40 years of space plant growth systems. *Life Sci Space Res.* 2016;10:1–16. <https://doi.org/10.1016/j.lssr.2016.06.004>.
19. Drew MC, Jackson MB, Giffard S. Ethylene-promoted adventitious rooting and development of cortical air spaces (aerenchyma) in roots may be adaptive responses to flooding in *Zea mays* L. *Planta.* 1979;147:83–8. <https://doi.org/10.1007/BF00384595>.
20. Sauter M, Steffens B. Biogenesis of Adventitious Roots and Their Involvement in the Adaptation to Oxygen Limitations. In: van Dongen JT, Licausi F, editors. *Low-Oxygen Stress in Plants: Oxygen Sensing and Adaptive Responses to Hypoxia*. Vienna: Springer; 2014. p. 299–312. https://doi.org/10.1007/978-3-7091-1254-0_15.
21. Dawood T, Yang X, Visser EJW, de Beek TAH, Kenschke PR, Cristescu SM, et al. A co-opted hormonal cascade activates dormant adventitious root primordia upon flooding in *Solanum dulcamara*. *Plant Physiol.* 2016;170:2351–64. <https://doi.org/10.1104/pp.15.00773>.
22. Steffens B, Rasmussen A. The physiology of adventitious roots. *Plant Physiol.* 2016;170:603–17. <https://doi.org/10.1104/pp.15.01360>.
23. Vidoz ML, Loreti E, Mensuali A, Alpi A, Perata P. Hormonal interplay during adventitious root formation in flooded tomato plants. *Plant J.* 2010;63:551–62. <https://doi.org/10.1111/j.1365-3113.2010.04262.x>.
24. Millar KDL, Johnson CM, Edelmann RE, Kiss JZ. An endogenous growth pattern of roots is revealed in seedlings grown in microgravity. *Astrobiology.* 2011;11:787–97. <https://doi.org/10.1089/ast.2011.0699>.
25. Wassink EC, Stolwijk JAJ. Effects of light quality on plant growth. *Annu Rev Plant Biol.* 1956;7:373–400. <https://doi.org/10.1146/annurev.pp.07.060156.002105>.
26. Paradiso R, Proietti S. Light-quality manipulation to control plant growth and photomorphogenesis in greenhouse horticulture: the state of the art and the opportunities of modern LED systems. *J Plant Growth Regul.* 2022;41:742–80. <https://doi.org/10.1007/s00344-021-10337-y>.
27. Ouzounis T, Rosenqvist E, Ottosen C-O. Spectral effects of artificial light on plant physiology and secondary metabolism: a review. *HortScience.* 2015. <https://doi.org/10.21273/HORTSCI.50.8.1128>.
28. Kiss JZ, Millar KDL, Edelmann RE. Phototropism of *Arabidopsis thaliana* in microgravity and fractional gravity on the International Space Station. *Planta.* 2012;236:635–45. <https://doi.org/10.1007/s00425-012-1633-y>.
29. Vandenbrink JP, Herranz R, Medina FJ, Edelmann RE, Kiss JZ. A novel blue-light phototropic response is revealed in roots of *Arabidopsis thaliana* in microgravity. *Planta.* 2016;244:1201–15. <https://doi.org/10.1007/s00425-016-2581-8>.
30. Camejo D, Frutos A, Mestre TC, del Carmen Piñero M, Rivero RM, Martínez V. Artificial light impacts the physical and nutritional quality of lettuce plants. *Hortic Environ Biotechnol.* 2020;61:69–82. <https://doi.org/10.1007/s13580-019-00191-z>.
31. Samuoliene G, Virsile A, Miliauskienė J, Haimi P, Laužikė K, Jankauskienė J, et al. The photosynthetic performance of red leaf lettuce under UV-A irradiation. *Agron.* 2020;10:761. <https://doi.org/10.3390/agronomy10060761>.
32. Massa GD, Newsham G, Hummerick ME, Morrow RC, Wheeler RM. Plant pillow preparation for the Veggie plant growth system on the International Space Station. *Gravit Space Res.* 2017;5:24–34. <https://doi.org/10.2478/gsr-2017-0002>.
33. Massa GD, Wheeler RM, Morrow RC, Levine HG. Growth chambers on the International Space Station for large plants. *Acta Hortic.* 2016:215–22. <https://doi.org/10.17660/actahortic.2016.1134.29>.
34. Overbey EG, Saravia-Butler AM, Zhang Z, Zea L, Costes SV, Galazka JM. NASA GeneLab RNA-seq consensus pipeline: Standardized processing of short-read RNA-seq data. 2023. <https://doi.org/10.1016/j.jisci.2021.102361>.
35. Krueger F, James F, Ewels P, Ayounian E, Weinstein M, Schuster-Boeckler B, et al. FelixKrueger/TrimGalore: v0.6.10 – add default decompression path. 2023. <https://doi.org/10.5281/zenodo.7598955>.
36. Dobin A, Davis CA, Schlesinger F, Drenkow J, Zaleski C, Jha S, et al. STAR: ultrafast universal RNA-seq aligner. *Bioinformatics.* 2013;29:15–21. <https://doi.org/10.1093/bioinformatics/bts635>.
37. Hosmani PS, Flores-Gonzalez M, Geest H van de, Maumus F, Bakker LV, Schijlen E, et al. An improved de novo assembly and annotation of the tomato reference genome using single-molecule sequencing, Hi-C proximity ligation and optical maps. 2019:767764. <https://doi.org/10.1101/767764>.
38. Fernandez-Pozo N, Menda N, Edwards JD, Saha S, Tecle IY, Strickler SR, et al. The Sol Genomics Network (SGN)—from genotype to phenotype to breeding. *Nucleic Acids Res.* 2015;43:D1036–41. <https://doi.org/10.1093/nar/gku1195>.
39. Li B, Dewey CN. RSEM: Accurate transcript quantification from RNA-seq data with or without a reference genome. *BMC Bioinformatics.* 2011;12:323. <https://doi.org/10.1186/1471-2105-12-323>.
40. Robinson MD, McCarthy DJ, Smyth GK. edgeR: a bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics.* 2009;26:139–40. <https://doi.org/10.1093/bioinformatics/btp616>.
41. Oliveros JC, Venny. An interactive tool for comparing lists with Venn's diagrams. <http://Bioinfogp.Cnb.Csic.Es/Tools/Venny/Index.html>. 2015:2015. <https://ci.nii.ac.jp/naid/20001505977>. Accessed 29 May 2023.
42. CRAN: cluster citation info. 2024. <https://cran.r-project.org/web/packages/cluster/citation.html>. Accessed 22 Dec 2024.
43. Rousseeuw PJ. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *J Comput Appl Math.* 1987;20:53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7).
44. Zhao S, Sun J, Shimizu K, Kadota K. Silhouette scores for arbitrary defined groups in gene expression data and insights into differential expression results. *Biol Proced Online.* 2018;20:5. <https://doi.org/10.1186/s12575-018-0067-8>.
45. Baker DN, Dyjack N, Braverman V, Hicks SC, Langmead B. Fast and memory-efficient scRNA-seq k-means clustering with various distances. *ACM BCB.* 2021;2021:24. <https://doi.org/10.1145/3459930.3469523>.
46. Wu T, Hu E, Xu S, Chen M, Guo P, Dai Z, et al. ClusterProfiler 4.0: a universal enrichment tool for interpreting omics data. *Innov (Camb).* 2021. <https://doi.org/10.1016/j.xinn.2021.100141>.
47. Xu S, Hu E, Cai Y, Xie Z, Luo X, Zhan L, et al. Using clusterProfiler to characterize multiomics data. *Nat Protoc.* 2024;19:3292–320. <https://doi.org/10.1038/s41596-024-01020-z>.
48. Rivera-Silva R, Chávez Montes RA, Jaimes-Miranda F. Gene ontology functional annotation datasets for the ITAG3.2 and ITAG4.0 tomato (*Solanum lycopersicum*) genome annotations. *Data Brief.* 2024;54:110401. <https://doi.org/10.1016/j.dib.2024.110401>.
49. Durinck S, Spellman PT, Birney E, Huber W. Mapping identifiers for the integration of genomic datasets with the R/Bioconductor package biomaRt. *Nat Protoc.* 2009;4:1184–91. <https://doi.org/10.1038/nprot.2009.97>.

50. Vandenbrink JP, Herranz R, Poehlman WL, Alex Feltus F, Villacampa A, Ciska M, et al. RNA-seq analyses of *Arabidopsis thaliana* seedlings after exposure to blue-light phototropic stimuli in microgravity. *Am J Bot*. 2019;106:1466–76. <https://doi.org/10.1002/ajb2.1384>.
51. Wu Q, Tong W, Zhao H, Ge R, Li R, Huang J, et al. Comparative transcriptomic analysis unveils the deep phylogeny and secondary metabolite evolution of 116 *Camellia* plants. *Plant J*. 2022;111:406–21. <https://doi.org/10.1111/tpj.15799>.
52. Olanrewaju GO, Haveman NJ, Naldrett MJ, Paul A-L, Ferl RJ, Wyatt SE. Integrative transcriptomics and proteomics profiling of *Arabidopsis thaliana* elucidates novel mechanisms underlying spaceflight adaptation. *Front Plant Sci*. 2023. <https://doi.org/10.3389/fpls.2023.1260429>.
53. Land ES, Sheppard J, Doherty CJ, Perera IY. Conserved plant transcriptional responses to microgravity from two consecutive spaceflight experiments. *Front Plant Sci*. 2024. <https://doi.org/10.3389/fpls.2023.1308713>.
54. Colla G, Roupael Y, Cardarelli M, Mazzucato A, Olimpieri I. Growth, yield and reproduction of dwarf tomato grown under simulated microgravity conditions. *Plant Biosystems - An International Journal Dealing with all Aspects of Plant Biology*. 2007;141:75–81. <https://doi.org/10.1080/11263500601153735>.
55. Li H, Lyu Y, Chen X, Wang C, Yao D, Ni S, et al. Exploration of the effect of blue light on functional metabolite accumulation in Longan embryonic calli via RNA sequencing. *Int J Mol Sci*. 2019;20:441. <https://doi.org/10.3390/ijms20020441>.
56. Guimba HDSS, Zhang X, Sierka E, Lin K, Ali MM, Ali WM, et al. Enhancement of photosynthesis efficiency and yield of strawberry (*Fragaria ananassa* Duch.) plants via LED systems. *Front Plant Sci*. 2022;13. <https://doi.org/10.3389/fpls.2022.918038>.
57. Liu M, Zhao Y, Fan P, Kong J, Wang Y, Xu X, et al. Grapevine plantlets respond to different monochromatic lights by tuning photosynthesis and carbon allocation. *Hortic Res*. 2023;10:uhad160. <https://doi.org/10.1093/hr/uhad160>.
58. Wakabayashi K, Soga K, Hoson T, Kotake T, Yamazaki T, Higashibata A, et al. Suppression of hydroxycinnamate network formation in cell walls of rice shoots grown under microgravity conditions in space. *PLoS ONE*. 2015;10:e0137992. <https://doi.org/10.1371/journal.pone.0137992>.
59. Han P, Shen S, Guo R, Zhao D, Lin Y-H, Jia S, et al. ROS is a factor regulating the increased polysaccharide production by light quality in the edible cyanobacterium *Nostoc flagelliforme*. *J Agric Food Chem*. 2019;67:2235–44. <https://doi.org/10.1021/acs.jafc.8b06176>.
60. Chen X, Cai W, Xia J, Yu H, Wang Q, Pang F, et al. Metabolomic and transcriptomic analyses reveal that blue light promotes chlorogenic acid synthesis in strawberry. *J Agric Food Chem*. 2020;68:12485–92. <https://doi.org/10.1021/acs.jafc.0c05020>.
61. Liu Z, Mao L, Yang B, Cui Q, Dai Y, Li X, et al. A multi-omics approach identifies *bHLH71-like* as a positive regulator of yellowing leaf pepper mutants exposed to high-intensity light. *Hortic Res*. 2023;10:uhad098. <https://doi.org/10.1093/hr/uhad098>.
62. Samkumar A, Jones D, Karppinen K, Dare AP, Sipari N, Espley RV, et al. Red and blue light treatments of ripening bilberry fruits reveal differences in signalling through abscisic acid-regulated anthocyanin biosynthesis. *Plant, Cell Environ*. 2021;44:3227–45. <https://doi.org/10.1111/pce.14158>.
63. Barker R, Kruse CPS, Johnson C, Saravia-Butler A, Fogle H, Chang H-S, et al. Meta-analysis of the space flight and microgravity response of the *Arabidopsis* plant transcriptome. *NPJ Microgravity*. 2023;9:21. <https://doi.org/10.1038/s41526-023-00247-6>.
64. Xu P, Chen H, Jin J, Cai W. Single-base resolution methylome analysis shows epigenetic changes in *Arabidopsis* seedlings exposed to microgravity space-flight conditions on board the SJ-10 recoverable satellite. *npj Microgravity*. 2018;4:1–11. <https://doi.org/10.1038/s41526-018-0046-z>.
65. Kruse CPS, Meyers AD, Basu P, Hutchinson S, Luesse DR, Wyatt SE. Spaceflight induces novel regulatory responses in *Arabidopsis* seedling as revealed by combined proteomic and transcriptomic analyses. *BMC Plant Biol*. 2020;20:237. <https://doi.org/10.1186/s12870-020-02392-6>.
66. NASA OSDR: OSD-217. <https://osdr.nasa.gov/bio/repo/data/studies/OSD-217>. Accessed 3 Jul 2025.
67. NASA OSDR: OSD-218. <https://osdr.nasa.gov/bio/repo/data/studies/OSD-218>. Accessed 3 Jul 2025.
68. Masson P, Su S, Galazka JM, Lai Polo S-H, Saravia-Butler AM, Boyko V, et al. Transcriptional profiling of roots and shoots from *Brachypodium distachyon* seedlings flown on the ISS. 2021. <https://doi.org/10.26030/2X6B-3V89>.
69. Ferl RJ, Zhou M, Strickland HF, Haveman NJ, Callahan JB, Bandla S, et al. Transcriptomic dynamics in the transition from ground to space are revealed by Virgin Galactic human-tended suborbital spaceflight. *NPJ Microgravity*. 2023;9:95. <https://doi.org/10.1038/s41526-023-00340-w>.
70. Paul A-L, Sng NJ, Zupanska AK, Krishnamurthy A, Schultz ER, Ferl RJ. Genetic dissection of the *Arabidopsis* spaceflight transcriptome: are some responses dispensable for the physiological adaptation of plants to spaceflight? *PLoS ONE*. 2017;12:e0180186. <https://doi.org/10.1371/journal.pone.0180186>.
71. Paul A-L, Haveman N, Califor B, Ferl RJ. Epigenomic regulators elongator complex subunit 2 and methyltransferase 1 differentially condition the spaceflight response in *Arabidopsis*. *Front Plant Sci*. 2021. <https://doi.org/10.3389/fpls.2021.691790>.
72. Paul A-L, Zupanska AK, Schultz ER, Ferl RJ. Organ-specific remodeling of the *Arabidopsis* transcriptome in response to spaceflight. *BMC Plant Biol*. 2013;13:112. <https://doi.org/10.1186/1471-2229-13-112>.
73. Nakashima J, Pattathil S, Avci U, Chin S, Alan Sparks J, Hahn MG, et al. Glycome profiling and immunohistochemistry uncover changes in cell walls of *Arabidopsis thaliana* roots during spaceflight. *NPJ Microgravity*. 2023;9:68. <https://doi.org/10.1038/s41526-023-00312-0>.
74. Manzano A, Carnero-Diaz E, Herranz R, Medina FJ. Recent transcriptomic studies to elucidate the plant adaptive response to spaceflight and to simulated space environments. *iScience*. 2022;25:104687. <https://doi.org/10.1016/j.isci.2022.104687>.
75. Choi W-G, Barker RJ, Kim S-H, Swanson SJ, Gilroy S. Variation in the transcriptome of different ecotypes of *Arabidopsis thaliana* reveals signatures of oxidative stress in plant responses to spaceflight. *Am J Bot*. 2019;106:123–36. <https://doi.org/10.1002/ajb2.1223>.
76. Beisel NS, Noble J, Barbazuk WB, Paul A-L, Ferl RJ. Spaceflight-induced alternative splicing during seedling development in *Arabidopsis thaliana*. *NPJ Microgravity*. 2019;5:9. <https://doi.org/10.1038/s41526-019-0070-7>.
77. Harrison K, Levy JG, Tamborindeguy C. Effects of 'Candidatus Liberibacter solanacearum' haplotypes A and B on tomato gene expression and geotropism. *BMC Plant Biol*. 2022;22:156. <https://doi.org/10.1186/s12870-022-03505-z>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.